Fisher forecasts from cached derivatives

How the derivatives are computed and stored, and why every FoM, error and shift is recomputed in seconds

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1 Summary

The question. The project's Fisher notebooks report figures of merit (FoMs), parameter errors and parameter shifts for the linear-bias and one-loop bias models, up to four linear-bias priors, seven scale cuts and four bin layouts, under dozens of assumed truths. One evaluation of the data vector – the model's prediction for every measured number, stacked in one list – runs CAMB, the code that computes the matter power spectrum, and CosmoLike, the code that projects it onto the sky for every thin bin; it takes seconds to tens of seconds. How, then, can a change of model, prior or scale cut cost only seconds?

The answer.

1. A Fisher forecast only needs the derivatives of the data vector at one point (Section 3).
2. The paper model is built from per-bin quantities – the biases, a counterterm and two stochastic amplitudes of every thin bin – tied to about twenty group-level parameters by calibrations, offsets and tilts (Section 4).
3. The base is one derivative column for each elementary knob: each cosmological parameter, each velocity dispersion, and each bias quantity in each thin bin. A model is a recipe that moves several knobs at once; written as a matrix M, it turns the base B into the model's derivatives, $\mathsf{D} = \mathsf{BM}$, and a new model is a new M (Section 5).
4. Each layout's base is computed once – by the notebook, or by an MPI script on many cores – and stored on disk (Section 6).
5. A scale cut keeps some rows of the data vector. Each cut is reduced once to a small matrix G (Section 7), and every Fisher matrix, FoM and shift is then a product of small matrices (Section 8).
6. A derivative needs more numerical accuracy than a χ^2, and a convergence study fixes the settings every published number uses (Section 9).
7. New evaluations are needed only for a term outside the base, a new fiducial point, or effects beyond first order (Section 10). Section 11 lists the files and when each must be recomputed.

parameter	meaning	fiducial	prior
$10^9 A_s$	primordial curvature amplitude	2.1	flat
n_s	scalar spectral index	0.96605	flat
H_0	Hubble constant [km/s/Mpc]	67.32	flat
Ω_b	baryon density today	0.04	flat
Ω_m	matter density today	0.3	flat
w_0	dark-energy equation of state today	-1	flat
w_a	its slope, $w(a) = w_0 + w_a(1-a)$	0	flat on $w_0 + w_a$
$\sum m_\nu$	neutrino mass sum [eV]	0.06	fixed in the Fisher
τ	optical depth to reionization	0.0697186	fixed

Table 1: The fiducial cosmology: the evaluate `override` of the paper yamls. The yamls' `ref` entries (chain starting points) differ from it and are not used.

The setup. The data are the DESI band powers C_ℓ^{gg}: the angular power spectrum of galaxy clustering – how strongly the galaxy counts in two directions on the sky correlate, as a function of the multipole ℓ, the angular frequency on the sky (the larger ℓ, the smaller the angle) – averaged in 53 bands of ℓ, one number per band: a band power. Each of six tracer groups g (BGS, LRG1, LRG2, LRG3, ELG1 and ELG2; a tracer is a galaxy sample that traces the matter) is sliced into thin bins, narrow adjacent slices in redshift, and a layout is one such slicing. Four layouts are used: bins of equal comoving width $\Delta d_{\rm com} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. The spectra are computed without the Limber approximation, which ties each multipole to a single wavenumber and fails for thin bins. Bin i has mean redshift $\bar{z}_i$, and $D(z)$ is the linear growth factor, $D(0) = 1$. The model being forecast is the paper model: a one-loop bias model with per-group offsets and tilts (defined in the higher-order note, `notes/higher_order_bias`) under the pessimistic linear-bias prior of the correlated-prior note, `notes/correlated_bias_prior_v2`, which allows correlated wiggles of the bin biases around a straight line in redshift (Section 4). The machinery lives in one notebook, `EXAMPLE_EVALUATE_FISHER_MODEL.ipynb` ("the notebook" below), on the Fisher module `interface/cosmolike_lssty1xdesi_fisher.py`; the script `scripts/compute_fisher_base_mpi.py` computes the same derivative base on MPI ranks (Section 6).

2 Fiducial model and parameter values

This section freezes the point at which every derivative of this note was taken, so the note does not depend on files that may drift.

The cosmology. The fiducial point is the `override` block – the entries that pin the evaluation point – of the paper yamls `EXAMPLE_EVALUATE1–4.yaml` (the runs' configuration files), the same in all four (Table 1). The Fisher matrices vary the first seven parameters; the neutrino mass stays at 0.06 eV and τ is fixed. At this point $\sigma_8 = 0.8278$ (the rms linear density contrast in spheres of $8\,h^{-1}$Mpc, a common amplitude measure).

The layouts. The four paper yamls $n = 1$ to 4 carry the layouts $\Delta d_{\rm com} = 62$ Mpc and $\Delta z =$ 0.02, 0.05, 0.2 (the datasets `lssty1xdesi_desi_dr3` and `..._dr1_dz0p02`, `dz0p05`, `dz0p20`):

76, 90, 36 and 10 thin bins in the six groups BGS, LRG1, LRG2, LRG3, ELG1 and ELG2, each spectrum in 53 band powers of width $\Delta\ell = 10$ from $\ell = 15$ to 544. Table 2 gives the sizes.

The bias fiducial. Bin i of group g has $b_{1,i} = B_g/D(\bar{z}_i)$, with D read off the linear power spectrum ratio at $k = 0.005$/Mpc and the group amplitudes

	BGS	LRG1	LRG2	LRG3	ELG1	ELG2
B_g	1.34	1.7	1.7	1.7	0.84	0.84

$b_{2,i}$ sits on the Lazeyras calibration [1], $b_2^{\rm L}(b_1) = 0.412 - 2.143\,b_1 + 0.929\,b_1^2 + 0.008\,b_1^3$, and $b_{s^2,i}$ on co-evolution, $-\frac{4}{7}(b_{1,i} - 1)$ (Section 4 defines both relations); at the fiducial every offset, tilt, bias residual, counterterm ($\alpha_{0,i}$, $\alpha_{2,i}$, both in $h^{-2}\,\mathrm{Mpc}^2$) and stochastic amplitude ($\mathrm{SN}_{0,i}$ in units of the bin's Poisson power $1/\bar{n}_i$; $\mathrm{SN}_{2,i}$ in units of $f_{\rm sat}\sigma_{\rm sat}^2/\bar{n}_i$, with $f_{\rm sat}\sigma_{\rm sat}^2 = 6.084$ for BGS and LRG and 0.841 for ELG, in $(\mathrm{Mpc}/h)^2$) is zero. The linear-bias prior is the pessimistic one: residual width $\sigma_g = 0.03$ in every group, correlation length $L = 0.05$ in redshift, charging – penalizing in the χ^2 – only the wiggles around each group's free line (Section 7).

The velocity dispersions. Seven σ_v groups (BGS split at $\bar{z} = 0.255$), each with a Gaussian prior centered on its fiducial; the paper widths are six times the exploration widths that the α_2-free exploration fits use:

	BGS0	BGS1	LRG1	LRG2	LRG3	ELG1	ELG2
fiducial [km/s]	371	358	357	348	343	304	289
paper width [km/s]	870	630	420	300	300	240	180
exploration width [km/s]	145	105	70	50	50	40	30

The yaml priors are truncated at $\sigma_v = 0$; at the paper width the BGS0 truncation is severe (the truncated prior's mean is about 850 km/s, not the fiducial 371). The Fisher matrices treat every σ_v prior as an untruncated Gaussian of the quoted width, centered on the fiducial; these widths are what the derivative caches store.

The model variants on this point. The paper model is the fiducial model: per-group offsets and tilts, $\alpha_2 = 0$ in every bin, the paper σ_v widths. The variants of the $k_{\rm max}$ scan fit it with α_2 free, with and without the tilts, on the exploration σ_v widths, with free $\alpha_{2,g}$ per group and the Gaussian widths 2 ($\Delta_{2,g}$), 1 ($\Delta_{s,g}$), 2 ($\mathrm{SN}_{0,g}$) and 5 ($\mathrm{SN}_{2,g}$); the counterterm widths are $0.5/k_{\rm max}^2$ at each cut, and every tilt is flat. All variants run under the baseline ($\sigma_g = 0.03$ for BGS, 0.01 otherwise; $L = 0.2$), pessimistic or free linear-bias prior.

The scale cuts. The non-Limber ultraviolet-fraction criterion of Saraivanov et al. [3] at ultraviolet fraction 0.05 (spelled out in Section 7), bin i cut at $k_{\rm max}/D(\bar{z}_i)$ with $k_{\rm max}$ quoted at $z = 0$, intersected with the shipped mask, the per-band keep flags of the data files ($k = 0.1185$/Mpc $= 0.176\,h$/Mpc in every bin). The cuts are computed on the paper fiducial above, at the final derivative settings (those of the derivative base), for $k_{\rm max} = 0.05, 0.075, \ldots, 0.25\,h$/Mpc, and cached in `chains/model_scale_cuts_final.npz` (Section 11); figures and FoMs use $k_{\rm max} \le 0.2\,h$/Mpc. Section 7 walks through what a cut does.

The covariance. Gaussian, computed at the one-loop fiducial above (with massless neutrinos, its one difference from the likelihood fiducial) for 14,000 square degrees; every Fisher matrix rescales its inverse by $f_A = 12{,}300/14{,}000 \approx 0.879$ to the survey area.

The derivative tier. Every number of this note is at the final tier, on every layout: CosmoLike `accuracyboost` 3, `nonlimber_accuracyboost` 64 (combined chi-grid factor $3 \times 64 = 192$, which the code rounds up to the power of two 256), `integration_accuracy` 1, `kmax_boltzmann` 20/Mpc (so the CAMB transfer reaches 60/Mpc), the low-rank engine's rank 5 and multipole cap 1000 (the non-Limber engine of Section 6.3), CAMB `AccuracyBoost` 2 and `k_per_logint` 30, with the halofit tolerance patch. Five-point stencils everywhere, never central differences: relative step $h = 0.02$ for the sampled parameters, per-bin operator steps 0.01 ($b_{1,i}$, relative), 0.02 ($b_{2,i}$ and $b_{s^2,i}$, absolute) and $5\,h^{-2}\,\mathrm{Mpc}^2$ ($\alpha_{0,i}$); the linear $\alpha_{2,i}$, $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ columns are exact (Section 6.3). Section 9 is the evidence.

3 Why derivatives at one point are enough

Let $m(p)$ be the expected data vector at the parameters p, and p_0 the fiducial point, the parameter values around which the forecast is made. The vector m has n entries: the band powers of every bin's auto spectrum (the bin correlated with itself) and of every measured cross spectrum between two bins, after 795 entries that the data format reserves for cosmic-shear blocks – the correlations of galaxy shapes distorted by lensing, not used here – and that stay zero for DESI alone. C is the covariance of the data: how the measured band powers scatter around m and how their errors correlate.

Near p_0 the data vector is close to a straight line in the parameters,

$$m(p) \;\simeq\; m(p_0) + \mathsf{D}\,(p - p_0), \qquad \mathsf{D}_{rj} = \left.\frac{\partial m_r}{\partial p_j}\right|_{p_0}, \tag{1}$$

with r one row, that is one band power.

With Gaussian data errors, the probability of measuring the data d when the parameters are p is proportional to $e^{-\chi^2/2}$, with $\chi^2(p) = [d - m(p)]^{\mathsf{T}}\,\mathsf{C}^{-1}\,[d - m(p)]$. Insert the straight line of Eq. (1): χ^2 becomes a quadratic in p, whose curvature – the matrix of its second derivatives, $2\,\mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{D}$ – no longer depends on the data. A prior is what is assumed about a parameter before the data arrive; a Gaussian prior multiplies the probability by a Gaussian of its own. The probability of the parameters given the data is then a Gaussian in p, whose inverse covariance is the Fisher matrix

$$F \;=\; \mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\mathsf{D} \;+\; F_{\mathrm{prior}}. \tag{2}$$

F_{prior} is the priors' contribution: a Gaussian prior of width w adds $1/w^2$ to its parameter's diagonal entry, and a flat prior, which prefers no value over another, adds nothing.

The inverse F^{-1} is the forecast covariance of the parameters. $\sigma_j = [(F^{-1})_{jj}]^{1/2}$ is the marginalized error of parameter j, its error when every other parameter is free. $\mathrm{FoM}(w_0, w_a) = [\det \mathsf{C}_w]^{-1/2}$ is the inverse area of the (w_0, w_a) error ellipse (a Gaussian's error ellipse has an area proportional to $\sqrt{\det \mathsf{C}_w}$), with C_w the (w_0, w_a) block of F^{-1} and $w(a) = w_0 + w_a(1 - a)$ the dark-energy equation of state (a the scale factor). $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ is defined the same

way, with Ω_m the matter density today in units of the critical density and A_s the amplitude of the primordial curvature power spectrum.

The model enters F only through D, at one point. The covariance is Gaussian, computed at the one-loop fiducial (the fiducial point of the one-loop bias model, Section 6.2) for 14,000 square degrees, with its inverse rescaled by $f_A = 12{,}300/14{,}000$ to the survey area: a Gaussian covariance is inversely proportional to the area (more sky, more independent modes, smaller errors), so the smaller survey multiplies the covariance by $1/f_A$ and its inverse by $f_A \approx 0.879$ (`invcov_scale` in the code and the caches). It does not depend on the model being fitted. Its own numerical accuracy was audited when it was built: its non-Limber distance grid sits where a finer grid moves the adjacent-bin cross band powers by less than 0.4%, while the next coarser grid misses them by up to 15–30% (`covariance/README.md` records the measured convergence).

4 The model being forecast

The quantities, bin by bin. Everything the notebooks fit is built from a small set of quantities:

- seven cosmological parameters, with flat priors: $10^9 A_s$, the spectral index n_s, the Hubble constant H_0, the baryon density Ω_b, Ω_m, w_0 and w_a;
- seven velocity dispersions $\sigma_{v,1}, \ldots, \sigma_{v,7}$, with Gaussian priors, one per set of bins that shares a value. They smooth each bin's redshift distribution for the random velocities of galaxies: a galaxy's measured redshift carries its line-of-sight velocity on top of the expansion, so random velocities blur where the bin's galaxies appear to sit (the Fingers-of-God effect). The sets follow the six tracer groups, except that BGS takes two: on $\Delta d_{\rm com} = 62$ Mpc the dispersions cover bins 1–10 and 11–18 (BGS), 19–29 (LRG1), 30–38 (LRG2), 39–49 (LRG3), 50–61 (ELG1) and 62–76 (ELG2);
- separately in every thin bin $i = 1, \ldots, N$: the linear bias $b_{1,i}$; the quadratic and tidal biases $b_{2,i}$ and $b_{s^2,i}$, the galaxies' responses to the squared density and to the tidal field; the counterterm $\alpha_{0,i}$, a term proportional to k^2 that absorbs small-scale physics; and the stochastic amplitudes $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$, noise-like terms from the discreteness of galaxies.

The per-bin power spectrum. The real-space galaxy power spectrum of bin i is

$$P_{gg,i}(k) \;=\; b_{1,i}^2\, P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) \;+\; \alpha_{0,i}\, k^2 P_{\rm mm}(k), \tag{3}$$

with $P_{\rm mm}$ the nonlinear matter power spectrum and $P_{\rm loop}$ the one-loop terms, the next order of perturbation theory beyond linear, which carry $b_{2,i}$ and $b_{s^2,i}$. $\alpha_{0,i}\, k^2 P_{\rm mm}$ is a counterterm, in $h^{-2}\,\mathrm{Mpc}^2$: a term that absorbs small-scale physics the model does not compute. In redshift space the model adds the stochastic terms $\mathrm{SN}_{0,i}$ (a white noise beyond the Poisson one) and $\mathrm{SN}_{2,i}\, k^2\mu^2$ (its line-of-sight counterpart, with μ the cosine of the angle to the line of sight). Section 6.3 writes out how every term reaches the projected band powers; the full physics is in the higher-order note, `notes/higher_order_bias`, and the correlated-prior note, `notes/correlated_bias_prior_v2`.

How the fit parameterizes the bins. No fit gives every bin its own free quantities; the parameter files tie the bins to their groups:

- the linear bias follows one amplitude B_g per group, $b_{1,i} = B_g/D(\bar{z}_i)$: the group's clustering amplitude $b_1 D$ is constant across its bins;
- $b_{2,i}$ follows $b_{1,i}$ through a calibration, the relation $b_2^{\rm L}(b_{1,i})$ fitted to simulations by Lazeyras et al. [1], plus one offset $\Delta_{2,g}$ per group; $b_{s^2,i}$ follows the co-evolution relation, the prediction for galaxies that form with no tidal bias and then move with the matter, $b_{s^2,i} = -\frac{4}{7}(b_{1,i} - 1) + \Delta_{s,g}$, with its own group offset;
- $\alpha_{0,i}$, $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ take one value per group: $\alpha_{0,g}$, $\mathrm{SN}_{0,g}$ and $\mathrm{SN}_{2,g}$;
- the tilts let a group value drift linearly in redshift: $\Delta_{2,g}$ becomes $\Delta_{2,g} + t_{2,g}(\bar{z}_i - \bar{z}_g)$, with $t_{2,g}$ the tilt and $\bar{z}_g$ the group's mean bin redshift, and likewise for $\Delta_{s,g}$ and $\alpha_{0,g}$; the stochastic amplitudes never tilt;
- the linear bias is fitted under the pessimistic prior of the correlated-prior note: one amplitude per bin, with a free amplitude and tilt per group and correlated residuals around them. (That note defines three more linear-bias priors, from growth-tied to free; the machinery below takes any of them.)

The paper model. The model this note forecasts is the paper model: the one-loop bias model above with, per group, the offsets $\Delta_{2,g}$ and $\Delta_{s,g}$, the counterterm amplitude $\alpha_{0,g}$, the stochastic amplitudes $\mathrm{SN}_{0,g}$ and $\mathrm{SN}_{2,g}$ and the three tilts, under the pessimistic linear-bias prior. Every group parameter carries the Gaussian prior width of Section 2, with all seven σ_v widths six times the exploration fits'. The widened dispersions stand in for the redshift-space counterterm α_2 of the DESI DR1 convention [2], a $k^2\mu^2$ term the model sets to zero in every bin: α_2 is about 99% degenerate with σ_v, and the widened σ_v absorbs a DESI-size α_2 at about the same FoM for $k_{\rm max} \geq 0.1\,h/\mathrm{Mpc}$ (the Fingers-of-God note, `notes/fog_parameter_degeneracy_study`). The same machinery builds the recipe matrix of any other model assembled from the same quantities, which the exploration notebooks fit and the correlated-prior and higher-order notes describe.

Why the forecast needs per-bin quantities. A chain samples only the group-level parameters above, about twenty in all. The forecast, though, must price choices that differ in exactly how the bins are tied together: one amplitude per group, per bin, or correlated in between; offsets with or without tilts. All of them are recipes on the same per-bin quantities, so the object worth precomputing is the response of the data vector to each quantity in each single bin. That is the base of the next section.

5 The base, and every model as a matrix on it

5.1 What the base is

The base is the list of elementary knobs the code turns by hand: each cosmological parameter, each velocity dispersion, and, separately in every thin bin, each of the six bias quantities of Section 4 plus the redshift-space counterterm $\alpha_{2,i}$ – seven per-bin families in all. The paper

layout	bins N	cross spectra	entries n	n_{base}	kept, shipped mask
$\Delta d_{\text{com}} = 62$ Mpc	76	107	10,494	547	5,674
$\Delta z = 0.02$	90	133	12,614	645	7,621
$\Delta z = 0.05$	36	52	5,459	267	2,982
$\Delta z = 0.2$	10	13	2,014	85	744

Table 2: The sizes of one layout's Fisher problem. Cross spectra: the pairs of bins whose cross spectrum is in the data vector. n: entries of the data vector. $n_{\text{base}} = 7N + 15$: columns of the base. Last column: the band powers the shipped mask keeps (the mask of the data files, before any scale cut), the largest set of rows any cut uses.

model fixes $\alpha_{2,i} = 0$, but the exploration fits free it and the shift tests of Section 8 assume truths with it, so the base carries its columns too. For each knob, the code computes how every band power changes when that knob alone moves: one derivative column, a vector of n numbers. These are the base columns, and they are what the notebook precomputes and stores (Section 8.1).

The base matrix B puts the base columns side by side. A model parameter, such as a group offset, a tilt or a per-bin amplitude, is a recipe that moves several knobs at once. The chain rule turns that recipe into a column of the model's derivative matrix: how m changes with p_j is the sum, over the knobs, of how m changes with the knob times how the knob changes with p_j,

$$\frac{\partial m}{\partial p_j} = \sum_a \frac{\partial m}{\partial \kappa_a}\,\frac{\partial \kappa_a}{\partial p_j}, \qquad \text{that is} \qquad \mathsf{D} = \mathsf{B}\,\mathsf{M}, \qquad \mathsf{M}_{aj} = \frac{\partial \kappa_a}{\partial p_j}, \tag{4}$$

with κ_a the value of knob a and p_j the model's parameter j. M is the recipes written as numbers: M_{aj} is how much knob a moves per unit of parameter j. Section 5.2 works one out.

There is no single b_1. Every bias knob belongs to one bin. "The b_1 columns" are therefore N columns, $\partial m/\partial b_{1,1}, \ldots, \partial m/\partial b_{1,N}$, and the same holds for each of the seven bias families $\{b_{1,i}\}$, $\{b_{2,i}\}$, $\{b_{s^2,i}\}$, $\{\alpha_{0,i}\}$, $\{\text{SN}_{0,i}\}$, $\{\text{SN}_{2,i}\}$ and $\{\alpha_{2,i}\}$, $i = 1, \ldots, N$. In order, B has

$$n_{\text{base}} = \underbrace{7}_{\text{cosmology}} + \underbrace{7}_{\sigma_v} + \underbrace{7N}_{7\text{ families}\times N\text{ bins}} + \underbrace{1}_{\text{check}} = 7N + 15 \tag{5}$$

columns; the check column is explained in Section 10. On $\Delta d_{\text{com}} = 62$ Mpc there are $N = 76$ bins, so B has $7 + 7 + 532 + 1 = 547$ columns of $n = 10{,}494$ numbers each. Table 2 gives every layout.

5.2 A toy example

The setup. Take one group g of $N = 3$ bins, with mean redshifts $\bar{z}_1$, $\bar{z}_2$, $\bar{z}_3$, their mean $\bar{z}_g$, and $D_i \equiv D(\bar{z}_i)$. Keep one band of each auto spectrum and of the two cross spectra of neighboring bins, so the data vector has $n = 5$ entries,

$$m = \left(C^{11}, C^{22}, C^{33}, C^{12}, C^{23}\right)^{\mathsf{T}}. \tag{6}$$

Keep one cosmological parameter θ and two bias families. The knobs are θ, $b_{1,1}$, $b_{1,2}$, $b_{1,3}$, $b_{2,1}$, $b_{2,2}$ and $b_{2,3}$, so $n_{\text{base}} = 7$.

The base matrix. Name the base columns $y = \partial m/\partial\theta$, $u_i = \partial m/\partial b_{1,i}$ and $v_i = \partial m/\partial b_{2,i}$. Then

$$\mathsf{B} = \begin{array}{c} \\ C^{11} \\ C^{22} \\ C^{33} \\ C^{12} \\ C^{23} \end{array} \begin{array}{c} \begin{array}{ccccccc} y & u_1 & u_2 & u_3 & v_1 & v_2 & v_3 \end{array} \\ \begin{pmatrix} * & * & 0 & 0 & * & 0 & 0 \\ * & 0 & * & 0 & 0 & * & 0 \\ * & 0 & 0 & * & 0 & 0 & * \\ * & * & * & 0 & * & * & 0 \\ * & 0 & * & * & 0 & * & * \end{pmatrix} \end{array}, \tag{7}$$

where $*$ marks a nonzero entry. The column u_1 vanishes on C^{22}, C^{33} and C^{23}, because the bias of bin 1 does not enter those spectra. A cross spectrum, such as C^{12}, depends on the biases of both its bins.

The model. The model has the parameters $p = (\theta, \Delta_{2,g}, t_{2,g}, \beta_1, \beta_2, \beta_3)$, so $n_{\mathrm{par}} = 6$, and sets the knobs by two recipes:

$$b_{1,i} = \frac{\beta_i}{D_i}, \qquad b_{2,i} = b_2^{\mathrm{L}}(b_{1,i}) + \Delta_{2,g} + t_{2,g}\,\zeta_i, \qquad \zeta_i \equiv \bar{z}_i - \bar{z}_g. \tag{8}$$

Here β_i is the amplitude of bin i, and b_2^{L} the calibration of Section 4, which predicts $b_{2,i}$ from $b_{1,i}$. $\Delta_{2,g}$ is the group's offset from that calibration and $t_{2,g}$ its tilt in redshift. Let $s_i \equiv b_2^{\mathrm{L}\prime}(b_{1,i})$ be the slope of the calibration.

The recipes as numbers. M_{aj} is how much knob a moves per unit of parameter j:

1. θ moves only θ. (y is computed by moving θ with the amplitudes β_i held fixed, which is how the model holds them, so θ needs no chain rule.)
2. $\Delta_{2,g}$ moves $b_{2,1}$, $b_{2,2}$ and $b_{2,3}$ by 1 each.
3. $t_{2,g}$ moves $b_{2,i}$ by ζ_i.
4. β_i moves $b_{1,i}$ by $1/D_i$ and, through the calibration, $b_{2,i}$ by s_i/D_i; the other bins do not move.

Each item is one column of

$$\mathsf{M} = \begin{array}{c} \\ \theta \\ b_{1,1} \\ b_{1,2} \\ b_{1,3} \\ b_{2,1} \\ b_{2,2} \\ b_{2,3} \end{array} \begin{array}{c} \begin{array}{cccccc} \theta & \Delta_{2,g} & t_{2,g} & \beta_1 & \beta_2 & \beta_3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/D_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/D_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/D_3 \\ 0 & 1 & \zeta_1 & s_1/D_1 & 0 & 0 \\ 0 & 1 & \zeta_2 & 0 & s_2/D_2 & 0 \\ 0 & 1 & \zeta_3 & 0 & 0 & s_3/D_3 \end{pmatrix} \end{array}. \tag{9}$$

The multiplication. Column j of $\mathsf{D} = \mathsf{BM}$ is the sum of the columns of B, weighted by column j of M:

$$\frac{\partial m}{\partial \theta} = y, \qquad \frac{\partial m}{\partial \Delta_{2,g}} = v_1 + v_2 + v_3,$$
$$\frac{\partial m}{\partial t_{2,g}} = \zeta_1 v_1 + \zeta_2 v_2 + \zeta_3 v_3, \qquad \frac{\partial m}{\partial \beta_i} = \frac{u_i + s_i v_i}{D_i}. \tag{10}$$

Written out entry by entry, the offset column is

$$\frac{\partial m}{\partial \Delta_{2,g}} = \left(\frac{\partial C^{11}}{\partial b_{2,1}}, \frac{\partial C^{22}}{\partial b_{2,2}}, \frac{\partial C^{33}}{\partial b_{2,3}}, \frac{\partial C^{12}}{\partial b_{2,1}} + \frac{\partial C^{12}}{\partial b_{2,2}}, \frac{\partial C^{23}}{\partial b_{2,2}} + \frac{\partial C^{23}}{\partial b_{2,3}} \right)^{\mathsf{T}} : \tag{11}$$

each auto spectrum responds to its own bin's $b_{2,i}$, and each cross spectrum to both of its bins'.

Another model, the same base. The free-per-bin model replaces the group offset and tilt by one offset per bin, $b_{2,i} = b_2^{\mathrm{L}}(b_{1,i}) + \Delta_{2,i}$. Each offset moves one knob, so its block on the $b_{2,i}$ rows is the 3×3 identity:

$$\mathsf{M}_{\text{free}} = \begin{array}{c} \\ \theta \\ b_{1,1} \\ b_{1,2} \\ b_{1,3} \\ b_{2,1} \\ b_{2,2} \\ b_{2,3} \end{array} \begin{array}{c} \begin{array}{ccccccc} \theta & \Delta_{2,1} & \Delta_{2,2} & \Delta_{2,3} & \beta_1 & \beta_2 & \beta_3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/D_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/D_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1/D_3 \\ 0 & 1 & 0 & 0 & s_1/D_1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & s_2/D_2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & s_3/D_3 \end{pmatrix} \end{array}, \qquad \frac{\partial m}{\partial \Delta_{2,i}} = v_i. \tag{12}$$

The priors.

- Group model: $1/w_g^2$ on $\Delta_{2,g}$ (a Gaussian prior of width w_g), nothing on $t_{2,g}$ (a flat prior).
- Free per bin: $1/w_g^2$ on each $\Delta_{2,i}$.
- Free per bin, $\sqrt{N}$-scaled: the same M_{free}, with $1/(3\,w_g^2)$ on each $\Delta_{2,i}$. The mean of the three offsets then has the width $w_g\sqrt{3}/\sqrt{3} = w_g$ of the group offset.

Three models, one B: the group and free models differ in M, the free model and its $\sqrt{N}$-scaled version only in F_{prior}. The base, the expensive part, never changes.

5.3 Every notebook model as a matrix on the base

The toy's recipes carry over to the paper model, with six bias families instead of two and six groups (BGS, LRG1, LRG2, LRG3, ELG1, ELG2) instead of one. The model is one matrix M of Eq. (4) on B, written down without any evaluation (`variant_columns` in the notebook). Figure 1 draws it. Its columns, in the figure's order:

- **Cosmology and σ_v** (θ_k, $\sigma_{v,k}$): unit vectors, as θ in the toy.

- **The group offsets and tilts** ($\Delta_{2,g}$, $t_{2,g}$, $\Delta_{s,g}$, $t_{s,g}$, $\alpha_{0,g}$ and its tilt, drawn for one group g; the ellipsis stands for the other groups and for $\mathrm{SN}_{0,g}$ and $\mathrm{SN}_{2,g}$, which never tilt). An offset puts ones on the rows of the group's bins in its family's block, as $\Delta_{2,g}$ in the toy; a tilt puts the weights $\bar{z}_i - \bar{z}_g$ on the same rows, as $t_{2,g}$, in every group of more than one bin, with a flat prior.

- **The per-bin amplitudes** ($\beta_1, \ldots, \beta_N$): the linear bias of each bin, $\beta_i = b_{1,i}\, D(\bar{z}_i)$. The paper model parameterizes $b_{1,i}$ through the group amplitude B_g, $b_{1,i} = B_g/D(\bar{z}_i)$, a free b_1 tilt per group, and the correlated residuals ϵ_i of the pessimistic prior; all three move the bins through these same columns, and the prior on the residuals enters F_{prior}, not M (Section 7). Through the calibrations of Section 4, $b_{2,i}$ follows $b_{1,i}$ with the slope $b_2^{\mathrm{L}\prime}$ and $b_{s^2,i}$ with $-4/7$, so the column has three entries:

$$\frac{\partial m}{\partial \beta_i} \;=\; \frac{1}{D(\bar{z}_i)} \left[\frac{\partial m}{\partial b_{1,i}} + b_2^{\mathrm{L}\prime}(b_{1,i})\, \frac{\partial m}{\partial b_{2,i}} - \frac{4}{7}\, \frac{\partial m}{\partial b_{s^2,i}} \right]. \tag{13}$$

Other models are other matrices M on the same base, built in the exploration notebooks (the correlated-prior note); switching needs no evaluation of the data vector.

On $\Delta d_{\mathrm{com}} = 62$ Mpc the paper model's M has $n_{\mathrm{par}} = 14 + 5\times 6 + 3\times 6 + 76 = 138$ columns: the cosmology and σ_v, five per-group families, the three tilt families, and one amplitude per bin.

6 Computing the base

The base columns are computed once per layout and cached on disk, by the notebook's section 4 or by the MPI script. This section is the machinery: the finite-difference stencil, the batched per-bin columns, the likelihood subclasses through which the steps reach CosmoLike, the exact columns of the linear terms, and what it all costs.

6.1 How a derivative column is computed

A finite difference. The code cannot differentiate CAMB and CosmoLike by hand. It moves one knob by a small step ϵ, recomputes the data vector, and divides the change by the step. The simplest such finite difference, the central difference $[m(\theta_0 + \epsilon) - m(\theta_0 - \epsilon)]/(2\epsilon)$, is off by a term of order ϵ^2 times the third derivative of m: the curvature it ignores.

The five-point stencil. A stencil is the set of points at which m is evaluated, with the weights that combine them. The code uses five points, $\theta_0 \pm \epsilon$, $\theta_0 \pm 2\epsilon$ and θ_0, with the weights

$$\frac{\partial m}{\partial \theta} \;\simeq\; \frac{m(\theta_0 - 2\epsilon) - 8\, m(\theta_0 - \epsilon) + 8\, m(\theta_0 + \epsilon) - m(\theta_0 + 2\epsilon)}{12\, \epsilon}. \tag{14}$$

These weights cancel the error terms through fourth order, so the remaining error falls as ϵ^4, and the formula is exact for any dependence up to quartic. The center point has weight zero and is never evaluated, so one column costs four evaluations.

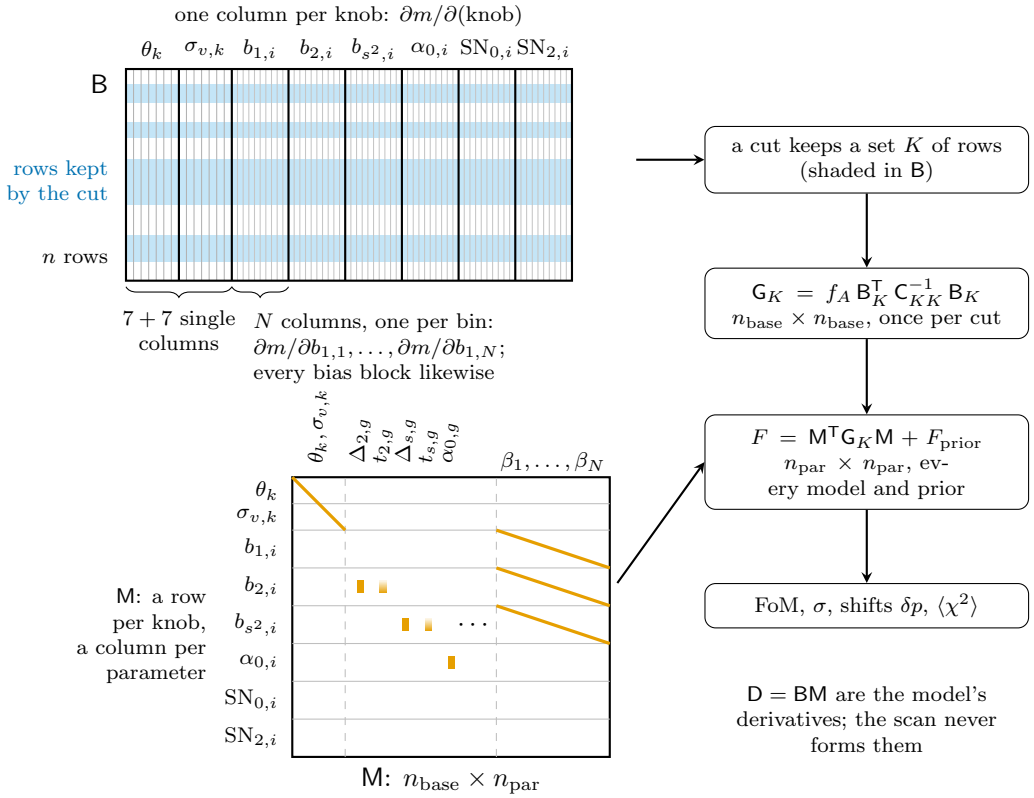


Figure 1: How the notebook assembles a Fisher matrix. Top left: the base B of one layout. Each cosmological parameter θ_k and each velocity dispersion $\sigma_{v,k}$ has one column; every bias family is a block of N columns, one per thin bin (thin gray lines). Shaded: the rows a cut keeps. Bottom left: a model's recipe matrix M, with a row per knob, in the column order of B, and a column per model parameter. Orange marks its nonzero entries: the identity on the cosmology and σ_v; for one group g, the offset $\Delta_{2,g}$ (ones on the group's bins in the $b_{2,i}$ block), its tilt $t_{2,g}$ (the weights $\bar{z}_i - \bar{z}_g$, drawn as a gradient), and likewise $\Delta_{s,g}$, $t_{s,g}$ and $\alpha_{0,g}$, the ellipsis standing for the other groups and the stochastic amplitudes; and three diagonals for the per-bin amplitudes β_i, Eq. (13). Right: a cut is reduced once to G_K, and every model and prior at that cut is $\mathsf{M}^{\mathsf{T}}\mathsf{G}_K\mathsf{M}$ plus priors; $f_A = 12{,}300/14{,}000$ rescales the inverse covariance to the survey area (Section 3). The $\alpha_{2,i}$ block and the check column of B are not drawn.

The cosmological and σ_v columns. These are stencils of the sampled parameters of the likelihood, taken by `model.derivatives` of the Fisher module (`interface/cosmolike_lssty1xdesi_fisher.py`). The step is relative, $\epsilon = h\,\theta_0$ with $h = 0.02$; a parameter whose fiducial value is zero and whose prior is flat (w_a, the one such case here) takes the absolute step h. No step depends on a prior width, which is why changing a prior never invalidates a cached column. Each evaluation is a full call of the likelihood, CAMB included, with the cache of previous results of cobaya (the framework that runs the likelihood) disabled, so that cobaya never hands back a vector saved at an earlier call in place of a fresh one (Appendix A).

The same call also returns the stencils of the six group amplitudes B_g, which set the linear bias of every bin of group g at the fiducial point, $b_{1,i} = B_g/D(\bar{z}_i)$. They enter no Fisher matrix

of the notebooks, because the per-bin columns replace them; they are kept for the chain-rule check of Section 10.

6.2 The per-bin columns, in batches

None of the four families $\{b_{1,i}\}$, $\{b_{2,i}\}$, $\{b_{s^2,i}\}$ and $\{\alpha_{0,i}\}$ of the galaxy power spectrum, Eq. (3), enters m linearly (Section 6.3), so each needs a stencil.

The stencil of one bin. For the linear bias of bin i, with every other knob at its fiducial value:

1. evaluate m four times, with $b_{1,i}$ replaced by $b_{1,i}(1 \pm \epsilon)$ and $b_{1,i}(1 \pm 2\epsilon)$;
2. combine the four vectors with the weights of Eq. (14), the step being $\epsilon\, b_{1,i}$:

$$\frac{\partial m}{\partial b_{1,i}} \simeq \frac{m\big(b_{1,i}(1-2\epsilon)\big) - 8\, m\big(b_{1,i}(1-\epsilon)\big) + 8\, m\big(b_{1,i}(1+\epsilon)\big) - m\big(b_{1,i}(1+2\epsilon)\big)}{12\,\epsilon\, b_{1,i}}. \quad (15)$$

The step of $b_{1,i}$ is relative, a fraction $\epsilon = 0.01$ of $b_{1,i}$. The steps of the other three are absolute: the four points are $b_{2,i} \pm \epsilon$ and $b_{2,i} \pm 2\epsilon$ with $\epsilon = 0.02$, the same for $b_{s^2,i}$, and $\epsilon = 5\,h^{-2}\,\mathrm{Mpc}^2$ for $\alpha_{0,i}$.

The problem: one bin at a time is wasteful. Done bin by bin, a family costs 4 evaluations per bin: $4 \times 76 = 304$ on $\Delta d_{\mathrm{com}} = 62$ Mpc, and 1,216 for the four families. Most of that work is wasted. The data vector of that layout holds 183 spectra, 76 autos and 107 crosses, and bin i enters at most six of them: its own auto spectrum and up to five cross spectra. Every other row of $\partial m/\partial b_{1,i}$ is exactly zero, and an evaluation that moves one bin leaves 177 or more spectra unchanged. Counted per spectrum, the block of one family, 183 spectra by 76 columns, is 98% zeros; in the cached arrays no column has a nonzero entry outside its own bin's spectra.

The idea: move several bins in one evaluation. Take five bins in a row whose only measured cross spectra join neighbors: 1–2, 2–3, 3–4 and 4–5. Put the bins in two batches, sets of bins moved together in one stencil:

- batch A moves $b_{1,i}$ in bins 1, 3 and 5 together;
- batch B moves $b_{1,i}$ in bins 2 and 4 together.

Each batch is one five-point stencil, four evaluations. Figure 2 shows batches on a real layout: each bin is a dot, a line joins two bins whose cross spectrum is in the data vector, and the number in each dot is the bin's batch. Table 3 shows which batch fills each entry of the five columns $\partial m/\partial b_{1,1}, \ldots, \partial m/\partial b_{1,5}$.

Reading the table:

- An auto spectrum C^{ii} is filled in the batch that moves bin i.
- A cross spectrum C^{ij} appears in two columns, filled from two different batches: $\partial C^{ij}/\partial b_{1,i}$ from the batch where bin i moves and bin j stays put, and $\partial C^{ij}/\partial b_{1,j}$ from the batch where j moves and i stays put.

spectrum	$\partial/\partial b_{1,1}$	$\partial/\partial b_{1,2}$	$\partial/\partial b_{1,3}$	$\partial/\partial b_{1,4}$	$\partial/\partial b_{1,5}$
C^{11}	A	0	0	0	0
C^{22}	0	B	0	0	0
C^{33}	0	0	A	0	0
C^{44}	0	0	0	B	0
C^{55}	0	0	0	0	A
C^{12}	A	B	0	0	0
C^{23}	0	B	A	0	0
C^{34}	0	0	A	B	0
C^{45}	0	0	0	B	A

Table 3: Five bins in a row, cross spectra between neighbors only. Each entry names the batch whose evaluations fill it: batch A moves bins 1, 3 and 5, batch B moves bins 2 and 4. A 0 is exactly zero, because the spectrum does not contain that bin.

- A 0 is not computed: a spectrum that does not contain bin i cannot depend on $b_{1,i}$, so the code sets those rows to zero.

Bin by bin, the five columns would cost $5 \times 4 = 20$ evaluations; the two batches cost $2 \times 4 = 8$.

A batch is a set of bins, not of spectra. No batch holds "the autos" or "the crosses". In the evaluations of the batch that contains bin i, bin i's auto spectrum and all of its cross spectra receive their derivative with respect to $b_{1,i}$. A cross spectrum C^{ij} is visited twice, once in the batch of each of its two bins. Figure 3c shows this on a real block.

When may two bins move together? Why not move every bin at once and keep, for each column, only its own rows?

- That works for the auto spectra: C^{ii} depends only on bin i's biases.
- It fails for the cross spectra. If bins i and j move together by δ, C^{ij} changes by $(\partial C^{ij}/\partial b_{1,i} + \partial C^{ij}/\partial b_{1,j})\,\delta$, and one evaluation cannot split the two terms.

So two bins may share a batch only if no measured spectrum contains both. Batching is "move as many bins at once as possible, then keep each column's own rows", with the fewest batches this rule allows.

What a batch is, as a picture. Draw each bin as a dot, and join two dots with a line when the data vector contains their cross spectrum: this is the cross-pair graph (Figure 2).

- A batch is a set of dots with no line between any two of them, so its bins can move together.
- Batching the bins means giving every dot a batch number, so that two joined dots never share a number.
- The code (`pair_coloring`) uses a greedy rule: it goes through the bins in data-vector order and gives each the lowest batch number that none of its already numbered neighbors has.

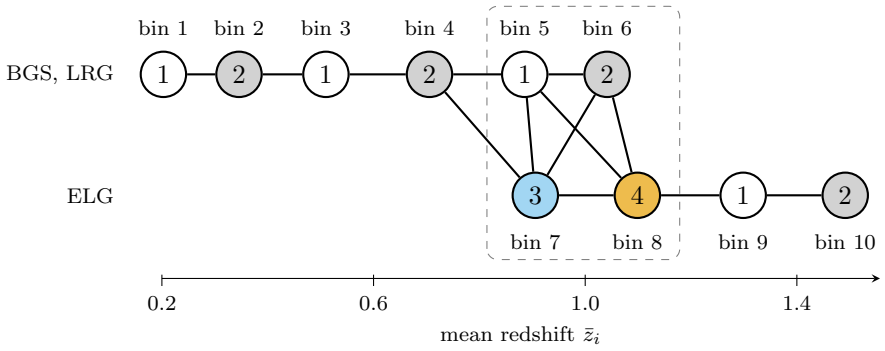


Figure 2: The cross-pair graph of the $\Delta z = 0.2$ layout, all ten bins. Each dot is a bin at its mean redshift (top row BGS and LRG, bottom row ELG), and a line joins two bins whose cross spectrum is in the data vector. The number in each dot is its batch from the greedy rule, repeated by the dot's shade. Bins 5 to 8 (dashed box: two LRG3 and two ELG1 bins, which overlap in redshift) are all joined to each other, so they need four different batches.

group	galaxies	bins	mean redshifts $\bar{z}_i$
BGS	bright galaxies	1–18	0.11–0.38
LRG1	luminous red galaxies	19–29	0.41–0.60
LRG2	luminous red galaxies	30–38	0.62–0.79
LRG3	luminous red galaxies	39–49	0.81–1.06
ELG1	emission-line galaxies	50–61	0.81–1.09
ELG2	emission-line galaxies	62–76	1.11–1.55

Table 4: The six groups of $\Delta d_{\rm com} = 62$ Mpc. LRG3 and ELG1 cover the same redshifts: they overlap.

Mathematicians call such batches colors, from the picture of labelling the dots so that joined dots never share a label; hence the name of the code's function.

The real layout, step by step. Take $\Delta d_{\rm com} = 62$ Mpc. Its $N = 76$ bins are numbered 1 to 76 in data-vector order, by group and, within a group, by redshift (numbers start at 1, as everywhere in this note and in Figure 3; the code counts from 0). DESI's galaxies come in six groups, ordered by redshift (Table 4). Figure 3 draws the whole layout and its batches; the steps below walk through it.

Which cross spectra are measured. The rule is simple: the data vector holds the cross spectrum of two bins when their redshift distributions $n(z)$ overlap or touch, that is, when the bins are neighbors in redshift. Bins far apart in redshift share no spectrum. For the fixed-Δz layouts this is the rule of `cross_pairs` in `scripts/make_desi_dz_case.py`. On $\Delta d_{\rm com} = 62$ Mpc the pair list is built by `scripts/make_desi_pairs.py`: it starts from the 97 pairs the DESI forecast measures and adds every pair whose kernels, blurred by the Fingers-of-God smoothing, still overlap, $O_{ij} \geq 10^{-6}$ with O_{ij} the normalized overlap of the two blurred $n(z)$. That adds ten LRG3–ELG1 pairs, for 107 in all. The threshold is not a tuning: any value between 10^{-6} and 0.0277 gives the same lists, and the blur adds no pair on the fixed-Δz layouts.

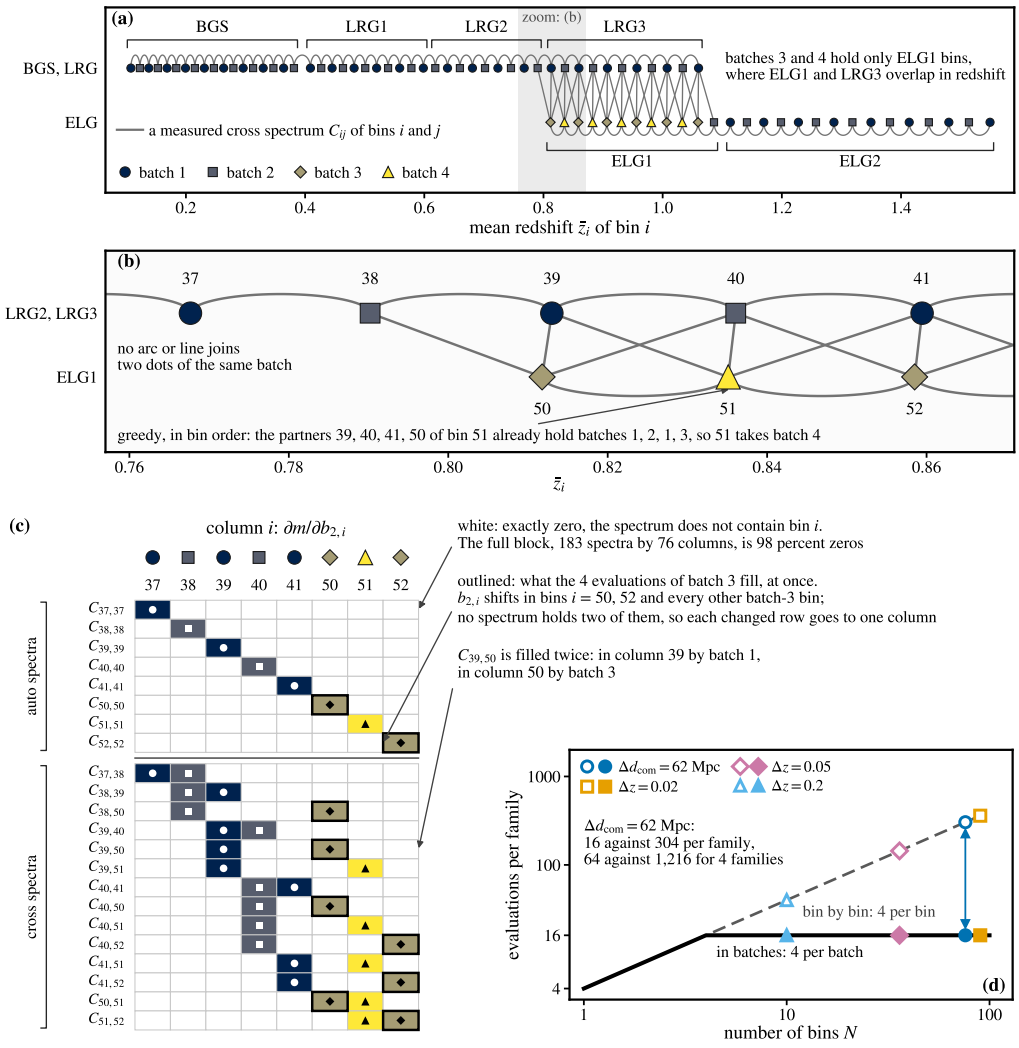


Figure 3: How the per-bin bias columns of $\Delta d_{\rm com} = 62$ Mpc are computed in batches. (a) Every bin at its mean redshift, BGS and LRG on the top row, ELG on the bottom; an arc or line is a measured cross spectrum, and the marker gives the batch. Look first at the long chains: each bin is crossed only with its neighbors, so the markers alternate between batch 1 (circle) and batch 2 (square). Then look at the overlap, where LRG3 and ELG1 cover the same redshifts: the lines between the rows are their cross spectra, and only there do batches 3 (diamond) and 4 (triangle) appear. (b) A zoom on the gray band of (a). No arc or line joins two markers of the same batch. Follow the greedy rule at bin 51: its earlier partners 39, 40, 41 and 50 already hold batches 1, 2, 1 and 3, so it takes batch 4. (c) The block $\partial m/\partial b_{2,i}$ of those eight bins, with a row per spectrum and a column per bin. A column is nonzero only on the spectra that contain its bin; white cells are exactly zero. The outlined cells are what the four evaluations of batch 3 fill at once. A cross spectrum such as $C_{39,50}$ is filled twice: in column 39 by batch 1, and in column 50 by batch 3. (d) Evaluations per bias family against the number of bins N: bin by bin the cost grows as $4N$ (open markers, dashed line); in batches it stops at $4 \times 4 = 16$ (filled markers), whatever N.

The cross-correlation note, `notes/cross_correlations`, derives the rule and the blur.

On $\Delta d_{\rm com} = 62$ Mpc the 107 cross spectra come in two kinds:

1. 74 join a bin to the next one along the chain of bins: neighboring thin shells, which touch at their shared edge. These include the four links at group boundaries: BGS–LRG1, LRG1–LRG2, LRG2–LRG3 and ELG1–ELG2.

2. 33 join an LRG bin to an ELG1 bin at nearly the same redshift: 32 LRG3–ELG1 pairs and one LRG2–ELG1 pair at $z \approx 0.8$. This is the overlap, explained below.

Each bin has 1 to 5 cross partners.

Along the chain: batches 1 and 2. The greedy rule walks through the bins in order. Bin 1 takes batch 1. Bin 2 is crossed with bin 1, so it takes batch 2. Bin 3 is crossed with bin 2 only, among the bins already numbered, so it takes batch 1 again, and so on: along the chain the bins alternate.

- The evaluations of batch 1 move bins $1, 3, 5, \ldots$ They fill $\partial C^{11}/\partial b_{1,1}$, $\partial C^{33}/\partial b_{1,3}$, $\ldots$, and the cross spectra C^{12}, C^{23}, C^{34} through their odd end: $\partial C^{12}/\partial b_{1,1}$, $\partial C^{23}/\partial b_{1,3}$, $\partial C^{34}/\partial b_{1,3}$.

- The evaluations of batch 2 move bins $2, 4, 6, \ldots$ and fill the even ends: $\partial C^{12}/\partial b_{1,2}$, $\partial C^{23}/\partial b_{1,2}$, $\partial C^{34}/\partial b_{1,4}$, and the even auto spectra.

This is the five-bin example of Table 3, continued through every group: the alternating circles and squares of Figure 3a.

The overlap: why batches 3 and 4 appear. LRG3 and ELG1 are two different galaxy populations, luminous red galaxies and emission-line galaxies, observed in the same volume, at $z \approx 0.8$–1.1 (Table 4). An LRG3 bin and an ELG1 bin at the same redshift trace the same matter, so their cross spectrum is measured, roughly $b_{1,i}\, b_{1,j}\, P$ with P the matter power spectrum. It carries no shot-noise term, because the two bins hold different galaxies: the $\mathrm{SN}_{0,i}$ cross terms exist only within one sample.

These pairs obey the same rule: the two bins are neighbors in redshift. They only look far apart because the bin numbers run group by group. LRG3 bin 44 ($\bar{z} = 0.932$) and ELG1 bin 55 ($\bar{z} = 0.931$) are 11 apart in number but at the same redshift, and their cross spectrum is measured. In Figure 2, drawn against redshift, the lines between the LRG and ELG rows are short.

For batching this matters because an ELG1 bin's redshift neighbors are its ELG neighbors and up to three LRG bins at nearly the same redshift. Bin 51, for example, is crossed with bins 50 and 52 (ELG1) and with bins 39, 40 and 41 (LRG3). Bins 39, 40 and 41 already sit in batches 1, 2 and 1, and bin 50 in batch 3, so bin 51 takes batch 4. The greedy rule puts the eleven ELG1 bins 50 to 60 in batches 3 (bins $50, 52, \ldots, 60$) and 4 (bins $51, 53, \ldots, 59$); batch 1 holds 33 bins and batch 2 holds 32, and the last ELG1 bin, 61, lands in batch 2. Batches 3 and 4 appear only in the overlap (Figure 3b).

Four batches, whatever the number of bins. Every bin's partners are local in redshift, so the number of batches does not grow with N. The other layouts follow the same pattern, neighbors along the chain plus overlap pairs: $\Delta z = 0.02$ has 133 cross spectra (88 along the chain) and 1 to 5 partners per bin; $\Delta z = 0.05$ has 52 (34) and 1 to 5; $\Delta z = 0.2$ has 13 (8) and 1 to 4. All four layouts get four batches.

A rule of thumb explains why the number stays small: if every bin's partners lie within k places of it in redshift order, $k + 1$ batches suffice, because going through the bins in that order, each bin has at most k numbered partners to avoid. The partners reach 3 places away on every layout. Four batches is also the minimum on every layout: each contains four bins all crossed with each other, two LRG3 and two ELG1 bins at the same redshift (bins 40, 41, 51 and 52 on $\Delta d_{\rm com} = 62$ Mpc; 36, 37, 51 and 52 on $\Delta z = 0.02$; 15, 16, 21 and 22 on $\Delta z = 0.05$; 5 to 8 on $\Delta z = 0.2$, Figure 2), and four bins all crossed with each other need four different batches.

On every layout batches 3 and 4 hold only ELG1 bins. On $\Delta d_{\rm com} = 62$ Mpc they are 11 of the 76 bins, about 15%, yet they take half of each family's evaluations.

What batching buys. The cost now follows from the number of batches, not from N. On $\Delta d_{\rm com} = 62$ Mpc:

1. one family takes 4 batches $\times$ 4 evaluations = 16 evaluations and fills all 76 of its columns, instead of $4 \times 76 = 304$;
2. the four stenciled families take $4 \times 16 = 64$ evaluations, instead of 1,216.

The same 64 hold on every layout, from 10 bins to 90 (Figure 3d).

Why not exact derivatives by hand? If each cross spectrum were a fixed bilinear expression in the biases, its derivatives could be written down exactly. The low-rank non-Limber engine of CosmoLike does not compute it that way. For each bin it approximates $\sqrt{P_{g,i}(k,z)}$, with $P_{g,i}$ the one-loop galaxy power spectrum of bin i, by a short sum of products of a function of k and a function of z: a truncated singular-value decomposition, which makes the projection fast.

It builds a cross spectrum from the geometric mean $\sqrt{P_{g,i}}\sqrt{P_{g,j}}$ of the two bins' spectra. Before the square root it floors $P_{g,i}$ at $10^{-15}\, b_{1,i}^2 P_\delta$, with P_δ the matter power spectrum, so the root always exists. The square root, the truncation and the floor make the spectra not exactly bilinear in the biases. Finite differences through the full likelihood differentiate the model exactly as it is computed.

How the steps reach CosmoLike. At every evaluation the likelihood computes all per-bin biases from the sampled parameters and hands them to CosmoLike through compiled setters. No sampled parameter moves one bin alone, so the base is evaluated on a subclass of each paper likelihood (`scripts/compute_fisher_base_mpi_classes.py`) that can add one per-bin step: it first sends the point's own arrays, exactly as its parent does, and then calls the stepped operator's setter once more with the step added to the batch's bins. CosmoLike thus receives a complete, consistent set of values and rebuilds only what depends on the ones that changed. Cobaya's cache of previous results is bypassed (`cached=False`), since every stencil evaluation is at the same sampled point. The code and the details, including what tests this, are in Appendix A.

term	route	depends on	linear in its amplitude?
density–density	low-rank (SVD, floor)	$b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$, $\alpha_{0,i}$; θ	no: stencils
redshift-space rows	FKEM split	$b_{1,i}$; f, D	no amplitude of its own
$\alpha_{2,i}$ counterterm	the same rows	$\alpha_{2,i}$; $b_{1,i}$, f	yes: a $\pm$ pair is exact
$\mathrm{SN}_{0,i}$, $\mathrm{SN}_{2,i}$	likelihood templates	$\mathrm{SN}_{0,i}$, $\mathrm{SN}_{2,i}$; W_i	yes: computed directly

Table 5: The terms of a DESI band power, Eqs. (17) and (21): the route each takes, what it depends on, and whether it is exactly linear in an amplitude of its own. θ stands for the cosmological parameters and the velocity dispersions σ_v.

6.3 Exact templates

Linear terms need no stencil. Suppose the data vector depends on an amplitude A only through one added term,

$$m(\theta, A) \;=\; m_{\mathrm{rest}}(\theta) \;+\; A\,T(\theta). \tag{16}$$

Then $\partial m/\partial A = T(\theta)$ exactly. No step enters, so there is no truncation error at all, not even the ϵ^4 error of the five-point stencil, Eq. (14). T, the term per unit amplitude, is called the *template* of A: one evaluation of the term, or the term computed directly from the fiducial kernels, gives the exact column. Three families enter m this way: the redshift-space counterterm $\{\alpha_{2,i}\}$ (below, under "Term 2"), and two whose templates the likelihood computes directly,

- the white noise beyond Poisson $\mathrm{SN}_{0,i}$, in units of the bin's Poisson power $1/\bar{n}_i$ ($\bar{n}_i$ its tracer density);
- the stochastic term $\mathrm{SN}_{2,i}\,k^2\mu^2$, with μ the cosine of the angle to the line of sight, in units of $f_{\mathrm{sat}}\sigma_{\mathrm{sat}}^2/\bar{n}_i$ (f_{sat} the satellite fraction, σ_{sat} the satellites' velocity dispersion in distance units).

That these are *exactly* linear is not obvious: the density–density part of every DESI spectrum goes through a singular value decomposition and a square root ("Why not exact derivatives by hand?"), and a square root is linear in nothing. The resolution is that none of the three families ever enters that route. To see it, write the whole spectrum once, term by term.

The whole C_ℓ, term by term. The pipeline evaluates the cross spectrum of bins i and j as a sum of two terms, each projected by its own route, and the likelihood adds the stochastic terms afterwards:

$$C_\ell^{ij} \;=\; \underbrace{C_\ell^{\mathrm{DD},\,ij}}_{\substack{\text{low-rank route:}\\ \text{SVD, geometric mean, floor}}} \;+\; \underbrace{C_\ell^{\mathrm{RSD},\,ij}}_{\substack{\text{FKEM split:}\\ \text{density}\times\text{RSD, RSD}\times\text{RSD}}} \;. \tag{17}$$

Every DESI row of the data vector is the average of C_ℓ^{ij} over one band of multipoles, and averaging is linear, so every linearity below survives into the band powers. Lensing magnification would be a fourth term; this forecast sets its amplitude to zero in every DESI bin. Table 5 summarizes the terms; the paragraphs below take them one at a time.

Term 1: density–density, the only term through the SVD. $C_\ell^{\rm DD}$ projects the real-space galaxy power $P_{gg,i}$ of Eq. (3). Per bin, the low-rank engine factorizes $\sqrt{P_{g,i}-P_{\infty,i}}$ by a singular value decomposition into five products of a function of k and a function of z; here $P_{\infty,i}=-(\sigma_4/2)\,D^4\,(b_{2,i}+2b_{s^2,i}/3)^2$ is the white-noise constant that the loop subtraction leaves at high k (σ_4 a fixed integral of the linear power), and a constant power projects exactly in Limber, so it is carried there instead. A cross pair uses the geometric mean $\sqrt{P_{g,i}}\sqrt{P_{g,j}}$, and a floor guards the square root. This term depends on $b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$ and $\alpha_{0,i}$ – and, through the square root, the rank truncation and the floor, on none of them exactly linearly.

Term 2: the redshift-space rows, through the FKEM split. With $W_i(\chi)$ the bin's kernel – its redshift distribution as a function of comoving distance χ, after the Fingers-of-God smoothing – D the growth factor, f the growth rate and j_ℓ the spherical Bessel function, the bin's linear redshift-space kernel is

$$\Delta_\ell^i(k) \;=\; \int d\chi\, W_i\, D\,\big[\, b_{1,i}\, j_\ell(k\chi) - f\, j_\ell''(k\chi)\,\big] \tag{18}$$

(each factor $f\mu^2$ of the flat-sky power becomes a $-j_\ell''$ in the exact projection; the summary note, `notes/bias_model_summary`, on the non-Limber spectra). Writing $\Delta_\ell^{i,\rm D}$ for the density part alone, the $b_{1,i}\,j_\ell$ term of Δ_ℓ^i, the term is

$$C_\ell^{{\rm RSD},\,ij} \;=\; \frac{2}{\pi}\int dk\,k^2\,P_{\rm lin}(k)\,\Big[\,\Delta_\ell^i\,\Delta_\ell^j - \Delta_\ell^{i,\rm D}\,\Delta_\ell^{j,\rm D}\,\Big] \;+\; \text{Limber corrections on the same rows,} \tag{19}$$

the density×RSD and RSD×RSD products, computed with the separable linear spectrum by FFTLog – a fast Fourier transform on a grid log-spaced in distance, which computes the Bessel-function integrals quickly (Section 9 describes the grid) – and corrected in Limber with the full one. This is the FKEM split, named for Fang, Krause, Eifler and MacCrann [4], whose method it is: the separable part exactly, the correction in Limber. The term depends on $b_{1,i}$, already covered by its stencil, and on cosmology through f and D. It has no amplitude of its own.

One amplitude does ride these rows: the redshift-space counterterm $\alpha_{2,i}$, which in the DESI DR1 convention [2] adds $(b_1+f\mu^2)\,f\,\alpha_{2,i}\,\mu^2k^2P_{\rm lin}$ to the bin's redshift-space power – a fixed term times $\alpha_{2,i}$, so the band powers are exactly linear in it and a single central difference, $[m(+\epsilon)-m(-\epsilon)]/(2\epsilon)$ at $\epsilon=5\,h^{-2}\,{\rm Mpc}^2$ through the full likelihood, is the exact column. The columns are taken in the batches of Section 6.2, two evaluations per batch, and the forward and backward differences check the linearity: they agree to at most 1.6×10^{-10} of the column over the four layouts. The paper model fixes $\alpha_{2,i}=0$; the columns serve the fits that free it and the shift tests of Section 8.

Terms 3 and 4: the stochastic templates, added outside CosmoLike. ${\rm SN}_{0,i}$ adds ${\rm SN}_{0,i}/\bar n_i$ to $P_{g,i}$, a constant in k. For a constant power S the k integral of the exact projection closes by the orthogonality of the Bessel functions, $\frac{2}{\pi}\int dk\,k^2\,j_\ell(k\chi_1)\,j_\ell(k\chi_2)=\delta_{\rm D}(\chi_1-\chi_2)/\chi_1^2$ ($\delta_{\rm D}$ the Dirac delta), so the spectrum gains, independently of ℓ,

$$C_\ell^{ij} \;\supset\; S\int d\chi\,\frac{W_i(\chi)\,W_j(\chi)}{\chi^2}, \qquad S=\frac{{\rm SN}_{0,i}}{\bar n_i}\ \text{ on the auto spectrum.} \tag{20}$$

A cross pair inside one tracer sample takes the mean of its two bins' stochastic powers, and a pair of two different samples takes none: different galaxies share no stochastic term. The integral is fixed by the fiducial kernels, so the band powers are exactly linear in $\mathrm{SN}_{0,i}$, with the integral as the template. The $\mathrm{SN}_{2,i}$ term is $S_2\,k^2\mu^2$ with $S_2 = \mathrm{SN}_{2,i}\,f_{\rm sat}\sigma^2_{\rm sat}/\bar{n}_i$; since $k^2\mu^2 = k^2_\parallel$ is the Fourier image of a second derivative along the line of sight, two integrations by parts move one derivative onto each kernel, and the template is $\int d\chi\, W_i' W_j'/\chi^2$ with $W' = dW/d\chi$ (the higher-order note, `notes/higher_order_bias`, on the stochastic terms in projection). The likelihood computes both templates directly from the fiducial kernels (`stochastic_templates` of `likelihood/desi_cl.py`, orders 0 and 2) and adds amplitude times template to the vector CosmoLike returned, keeping the sum in `last_datavector`. The amplitudes never enter CosmoLike's projection at all, let alone the SVD.

The consequence. Collecting the terms, the DESI part of the data vector is

$$m \;=\; m_{\rm DD+RSD}\big(\{b_{1,i}\},\{b_{2,i}\},\{b_{s^2,i}\},\{\alpha_{0,i}\};\,\theta\big) + \sum_i \alpha_{2,i}\,T_i^{(a_2)} + \sum_i \mathrm{SN}_{0,i}\,T_i^{(0)} + \sum_i \mathrm{SN}_{2,i}\,T_i^{(2)}, \tag{21}$$

with θ the cosmological parameters and the σ_v. The last three groups have the form of Eq. (16), so their columns are exact – the stochastic templates computed directly, the α_2 columns by one central difference each; only the first group needs stencils.

Why $\alpha_{0,i}$ is not a template. It looks like one: a counterterm amplitude multiplying $k^2 P_{\rm mm}$ in Eq. (3). But it sits in the real-space density–density power, the one term that goes under the square root, the SVD and the geometric mean, so the projected spectra are not exactly linear in it; the same holds for $b_{1,i}$, $b_{2,i}$ and $b_{s^2,i}$. That is the dividing line of this section: the families that stay outside the density–density term get exact templates, and everything that passes through the SVD gets the stencils of Section 6.2.

The templates themselves are validated in the higher-order note, `notes/higher_order_bias`, section "Validation": the exact $\mathrm{SN}_{2,i}$ projection against a brute-force j_ℓ, j_ℓ'' double integral, and the linearity of the $\mathrm{SN}_{2,i}$ term by forward against backward differences.

6.4 How many data-vector evaluations, and how long

Counted in the notebook and the Fisher module, the base of one layout takes

$$N_{\rm eval} \;=\; 1 + 4\,N_{\rm batch}\,N_{\rm fam} + 4\,N_{\rm stencil} + 2\,N_{\rm batch} + 1 \tag{22}$$

evaluations of the data vector, term by term in Table 6. It is 154 on every layout, because no term grows with the number of bins. Section 6.5 counts the evaluations when the pair list grows and more batches are needed.

The cost of one evaluation. Each evaluation is a complete data vector: CAMB, then CosmoLike's non-Limber projection of every thin bin. Its cost depends mostly on g, the refinement factor of the comoving-distance grid on which the non-Limber projection integrates (Section 9). Table 7 gives the times the convergence study measured on $\Delta z = 0.05$ (36 bins), on AC power. The MCMC settings are those of the example yamls used for Markov-chain Monte Carlo runs, the sampling of the full likelihood.

term	what it is	evaluations
1	the fiducial vector, templates and biases	1
$4\,N_{\rm batch}\,N_{\rm fam}$	4 batches, 4 stenciled families $\{b_{1,i}\}$, $\{b_{2,i}\}$, $\{b_{s^2,i}\}$, $\{\alpha_{0,i}\}$	64
$4\,N_{\rm stencil}$	20 stencil parameters: 7 cosmology, 7 σ_v, 6 B_g	80
$2\,N_{\rm batch}$	the exact $\{\alpha_{2,i}\}$ columns: 4 batches, + and −	8
1	the fiducial point again, after every step (the reset check)	1
	total	154

Table 6: The evaluations behind one layout's base, in the order they run. The $\{\mathrm{SN}_{0,i}\}$ and $\{\mathrm{SN}_{2,i}\}$ templates cost no evaluation beyond the fiducial one, whose kernels they are read from.

configuration	grid factor g	seconds per evaluation
MCMC settings	32	3.9
starting configuration, grid lowered	64	11.4
starting configuration	128	17.0
starting configuration, grid raised	256 (final)	28.0
starting configuration, grid raised	512	48.4
final settings	256 (final)	19.5

Table 7: The cost of one data-vector evaluation on $\Delta z = 0.05$, measured by the convergence study on a MacBook Pro with the Apple M2 Pro chip (8 OpenMP threads, one BLAS thread), on AC power with no other heavy job running, except the $g = 256$ run of the starting configuration, which started while one other job was running. The final settings ran 130 evaluations in 42 minutes. Section 9 describes the configurations.

The total time. The base of a layout takes $N_{\rm eval} \times t_{\rm eval}$, with $t_{\rm eval}$ the time of one evaluation, measured on a MacBook Pro with the Apple M2 Pro chip (8 OpenMP threads, one BLAS thread, on AC power with no other heavy job). At the final settings:

- $\Delta z = 0.05$, from the measured $t_{\rm eval}$: 154×19.5 s ≈ 50 minutes.
- The other layouts, a rough estimate that scales $t_{\rm eval}$ with the number of bins: $\Delta d_{\rm com} = 62$ Mpc (76 bins) 154×41 s ≈ 1.8 hours; $\Delta z = 0.02$ (90 bins) 154×49 s ≈ 2.1 hours; $\Delta z = 0.2$ (10 bins) 154×5.4 s ≈ 14 minutes.

The four layouts then take about five hours. On other hardware – an Intel workstation, an x86 cluster node – the time per evaluation differs, so these numbers are a guide to the order of magnitude, not a benchmark.

The same base on MPI ranks. `scripts/compute_fisher_base_mpi.py` splits the 154 evaluations of one layout over N MPI ranks, each running CosmoLike's and CAMB's OpenMP loops on its own threads, and writes the same two caches. The evaluations are independent, so rank 0 assembles the gathered vectors with the serial arithmetic; on one rank the script evaluates in the notebook's own order, and on $\Delta z = 0.2$ the 1×8 and 2×4 (ranks $\times$ threads) runs reproduce the serial caches bit for bit. Each rank holds a complete model, covariance included, and peaks at about 10 to 15 GB of memory on $\Delta d_{\rm com} = 62$ Mpc, which sets how many ranks a node fits. The project README, section "Computing the derivative base with

MPI", gives the launch commands; the script's docstring holds the cache rules and the parallel layout.

6.5 More cross spectra: how many batches, and at what cost

Every count above assumes the current pair lists. Two changes lengthen them: measuring the spectra of bins two apart within a sample, $C^{i,i+2}$, and rebuilding the lists at the larger dispersions σ_v that the paper priors allow (both below; `notes/cross_correlations` forecasts what those pairs add). The price of a longer list is set by Eq. (22): with $N_{\rm fam} = 4$ and $N_{\rm stencil} = 20$ fixed,

$$N_{\rm eval} \;=\; 1 + 16\,N_{\rm batch} + 80 + 2\,N_{\rm batch} + 1 \;=\; 82 + 18\,N_{\rm batch},$$

so each extra batch costs 18 evaluations: 154, 172 and 190 for 4, 5 and 6 batches. This subsection counts the batches a longer list needs, shows that the code's greedy rule can use more than the minimum there, and names the fix.

The rule, replayed on a toy. Two bins may share a batch only if no measured spectrum contains both (Section 6.2). Take six bins in a row. When only neighbors are measured, $C^{i,i+1}$ for $i = 1, \ldots, 5$, odd and even bins alternate and two batches suffice. Now also measure $C^{i,i+2}$ for $i = 1, \ldots, 4$. Bins 1 and 3 can no longer share a batch, because the measured C^{13} contains both; neither can bins 2 and 4: the odd/even split fails. Three batches work, by giving bin i the number $1 + \big((i-1) \bmod 3\big)$: two bins with the same number are then 3 or 6 places apart, every measured spectrum joins bins at most 2 apart, so no spectrum contains two bins of one batch.

bin	1	2	3	4	5	6
batch, $C^{i,i+1}$ measured	1	2	1	2	1	2
batch, $C^{i,i+1}$ and $C^{i,i+2}$ measured	1	2	3	1	2	3

Three is also the fewest: bins 1, 2 and 3 are measured pairwise (C^{12}, C^{13}, C^{23}), so each of the three pairs must be split and three different numbers are forced. In general, when every measured pair lies at most b places apart – the list has *bandwidth* b – the pattern $i \bmod (b+1)$ colors it with $b+1$ batches; this is the rule of thumb of Section 6.2 again.

The floor: cliques. On the cross-pair graph (bins are dots, measured pairs are lines, a batch is a color; Section 6.2) the toy's second argument generalizes. A *clique* is a set of bins every two of which share a measured spectrum – a set of dots with a line between every two. Any two bins of a clique must sit in different batches, so a clique of q bins forces at least q batches, whatever the rest of the layout looks like: the largest clique is a floor on $N_{\rm batch}$. The floor is not always reachable – some graphs need more colors than their largest clique – so every list below is also checked by an exact search.

Today's lists sit on the floor. On every layout the largest clique has four bins: two adjacent LRG3 bins and the two ELG1 bins they overlap, the quadruplets of Section 6.2 (bins 40, 41, 51 and 52 on $\Delta d_{\rm com} = 62$ Mpc). The four layouts hold 10, 14, 5 and 1 such four-cliques, every one of the form LRG3–LRG3–ELG1–ELG1, and no clique of five bins exists. The greedy rule colors every layout with four batches, so on today's lists the code is optimal: four is both what it uses and the fewest possible, and $N_{\rm eval} = 154$.

layout	pairs today	pairs	clique	greedy	DSATUR = min	$N_{\rm eval}$ greedy / min
$\Delta d_{\rm com} = 62$ Mpc	107	177	4	6	4	190 / 154
$\Delta z = 0.02$	133	217	4	6	4	190 / 154
$\Delta z = 0.05$	52	82	4	6	4	190 / 154
$\Delta z = 0.2$	13	17	4	4	4	154 / 154

Table 8: The pair lists with every within-sample $C^{i,i+2}$ added to today's lists, per layout: the number of cross pairs, the largest clique, the batches of the greedy rule in bin order, the batches of DSATUR (equal to the exact minimum on every layout), and $N_{\rm eval} = 82 + 18\,N_{\rm batch}$ for both colorings.

Measuring every $C^{i,i+2}$: the floor stays at four. Suppose every spectrum of bins two apart *within a sample* is measured – bandwidth 2 inside BGS, LRG and ELG, where a sample is the galaxy population, so a pair two apart may straddle a group boundary – while the cross-sample pairs stay today's. That adds 70, 84, 30 and 4 pairs on the four layouts (Table 8). The largest clique does not grow: a new pair joins two bins of one sample, and within one sample the largest clique is three consecutive bins (bins i and $i + 3$ are not measured), so the LRG3–ELG1 quadruplets remain the largest; the exact search confirms that four batches still color every layout. Within one sample alone, with no cross-sample pairs, the toy's pattern i mod 3 would even do with three.

The greedy rule overshoots the floor. `pair_coloring` walks the bins in data-vector order, so it colors every LRG bin before any ELG1 bin. Along a bandwidth-2 chain each bin must avoid its two predecessors, so the LRG bins cycle through batches 1, 2, 3. Each ELG1 bin then arrives with up to five already-colored partners – its two ELG1 predecessors and up to three LRG3 bins at its redshift – and the LRG3 partners, cycling through 1, 2 and 3, can hold all three low numbers at once, so the ELG1 bins open batches 4, 5 and 6. The result is six batches on $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$ and $\Delta z = 0.05$ ($N_{\rm eval} = 190$, 36 evaluations above the floor) and four on $\Delta z = 0.2$, whose two ELG1 bins see fewer partners.

Figure 4 draws this on every layout. The bins with the most partners sit in the shared LRG3–ELG1 volume: seven partners each on the three thin layouts (the bin's two neighbours on each side in its own sample, plus three bins of the other sample). What sets the greedy count is not that total but the number of partners already colored when greedy reaches the bin. An LRG3 bin is reached before every ELG1 bin, so only its two predecessors count, and it lands in batch 1, 2 or 3; an ELG1 bin is reached after all of them, with up to five colored partners, and needs batch 4, 5 or 6. Greedy therefore uses at most one batch more than the largest number of earlier partners – five, hence six batches – not one more than the largest number of partners, which would allow eight.

A four-batch coloring exists on every layout, and the exact search produces one: inside each sample it cycles with period 3, the toy's pattern, and in the LRG3–ELG1 shared volume both groups cycle with period 4, out of phase, so that an LRG3 bin and the up-to-three ELG1 bins at its redshift carry four different numbers. On $\Delta d_{\rm com} = 62$ Mpc, bin by bin and split by group:

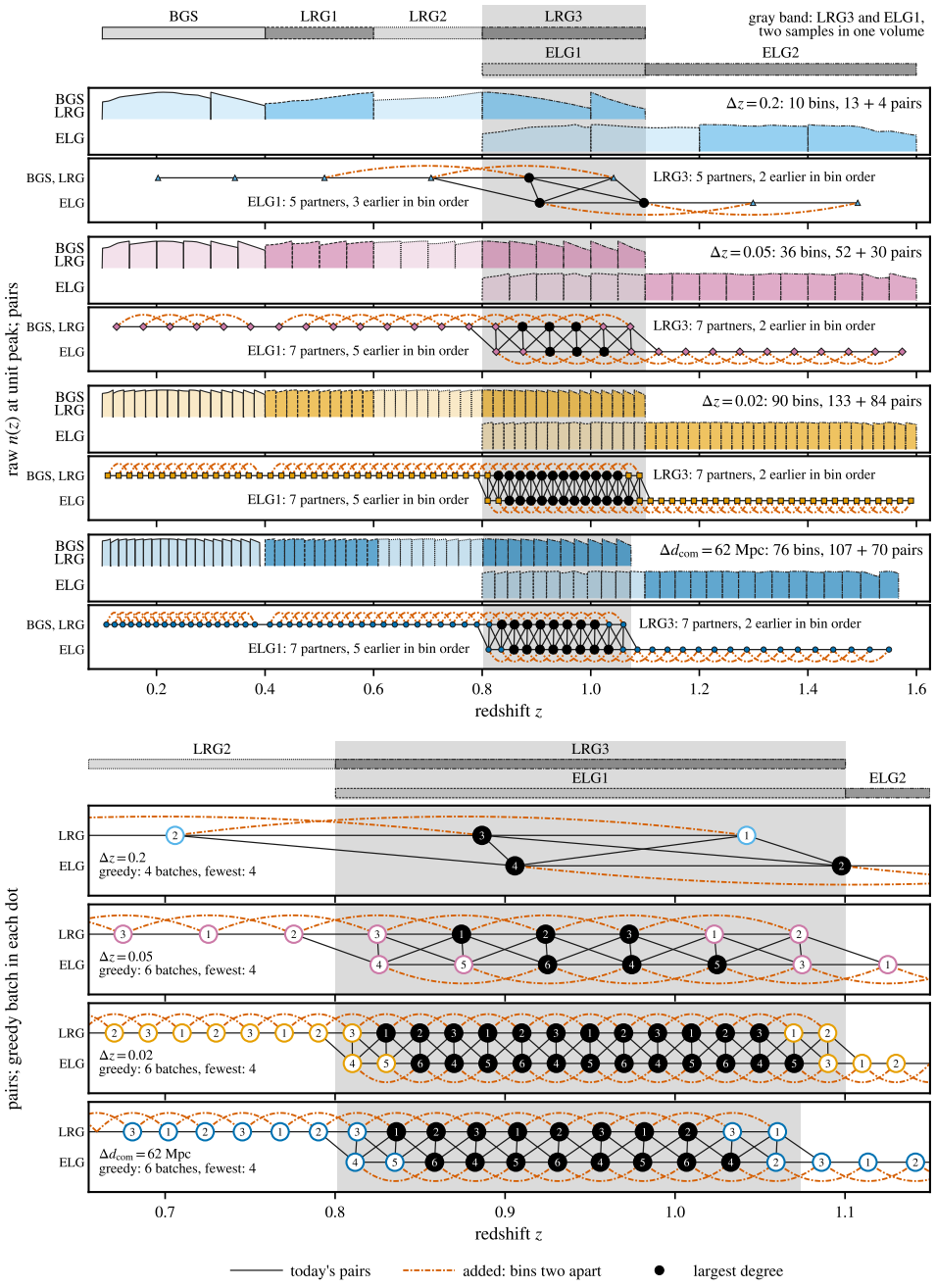


Figure 4: The pair graph of the four DESI layouts with every pair of bins two apart in one sample, $C^{i,i+2}$, added to today's cross pairs. Upper part: the raw $n(z)$ of each layout at unit peak and its pair graph; thin dark segments are today's pairs, vermillion dash-dotted arcs the added ones, drawn above the BGS/LRG row and below the ELG row. Black dots mark the bins with the most partners; the texts give that number and how many of those partners come earlier in bin order. Lower part: the LRG3/ELG1 volume magnified, with each bin's batch in the greedy coloring in bin order written in its dot (LRG3 cycles through 1–3, ELG1 through 4–6). The gray band marks where LRG3 and ELG1 observe the same volume.

group	greedy (six batches)	exact (four batches)
BGS	123123123123123123	231231231231231231
LRG1	12312312312	23123123123
LRG2	312312312	123123123
LRG3	31231231231	41234123412
ELG1	456456456423	234123412341
ELG2	123123123123123	231231231231231

One digit per bin, in bin order. In the greedy coloring the batches 4–6 live entirely in ELG1; in the exact coloring every group cycles with period 3 except LRG3 and ELG1, which cycle with period 4.

DSATUR, the standard fix. The overshoot is an artifact of the coloring order, not of the batching idea, and the standard cure is to choose the order on the fly. DSATUR ("degree of saturation") is the same greedy rule with one change: at each step it colors next the bin whose already-colored partners show the most *distinct* batch numbers, breaking ties by the number of partners. The bins under the most pressure are colored early, while the low numbers around them are still free. On every list of this subsection DSATUR finds a four-batch coloring in milliseconds, matching the exact search (Table 8). Coloring in mean-redshift order instead of bin order also gives four batches on every bandwidth-2 list, but five on the $\Delta z = 0.02$ list of case (a) below, so it is the weaker fix.

When would these pairs be measured? The pair rule of Section 6.2 keeps a cross spectrum when the two bins' Fingers-of-God-blurred kernels overlap, $O_{ij} \geq 10^{-6}$, evaluated at the fiducial dispersions. A larger σ_v blurs every kernel further, and the pairs of bins two apart – kernels separated by one full bin – are the first unlisted pairs to approach the threshold. The paper's widened σ_v priors leave room for that: the Fingers-of-God note, `notes/fog_parameter_degeneracy_study`, section on the pair list at the edges of the σ_v prior, finds the crossing at 0.34 and 0.52 of the widened widths on $\Delta d_{\rm com} = 62$ Mpc and $\Delta z = 0.02$, inside one sigma of the adopted prior.

How far does the fit itself go? The paper model carries no $\alpha_{2,i}$, so when the truth does, the fit absorbs it, mostly into the dispersions, and a negative α_2 raises them. The linear response – the first-order shift formula, Eq. (25) of Section 8 – gives the dispersion each group reaches under a truth $\alpha_{2,i} = -A$ in every bin, for three sizes: (a) $A = 0.5/k_{\rm max}^2\, h^{-2}\,{\rm Mpc}^2$, the counterterm prior width at the cut (Section 2); (b) $A = 12.5\, h^{-2}\,{\rm Mpc}^2$, the DESI-size truth of the shift tests; (c) $A = 25\, h^{-2}\,{\rm Mpc}^2$, twice that. Table 9 shows $\sigma_v^{\rm max}$, each group's largest dispersion over the cuts $k_{\rm max} = 0.05$–$0.2\, h/{\rm Mpc}$ (quoted at $z = 0$, as always).

The dispersions stay modest, 301–513 km/s, the largest being ELG1 and ELG2 under (a), where the response and the prior width $0.5/k_{\rm max}^2$ peak together at $k_{\rm max} = 0.1$–$0.125\, h/{\rm Mpc}$. They sit far below the roughly 900 km/s scale of BGS0's widened prior – width 870 km/s, truncated mean about 850 km/s (Section 2) – that the prior-edge crossing above involves. Rebuilt at these dispersions, the pair lists barely move: only $\Delta z = 0.02$ gains pairs – 28 under (a), all pairs of ELG bins two apart with overlaps 1.1×10^{-6} to 6.3×10^{-5}, which is 50 to 3000 times weaker than any pair on today's list of that layout, and 14 under (c) – while (b) adds nothing anywhere and the other three layouts gain nothing under any truth. On the 28-pair list the largest clique and the minimum stay at four batches; the greedy rule needs five

layout	truth	BGS0	BGS1	LRG1	LRG2	LRG3	ELG1	ELG2
	fiducial σ_v	371	358	357	348	343	304	289
	paper prior width	870	630	420	300	300	240	180
$\Delta d_{\rm com} = 62$ Mpc	(a)	469	456	449	443	439	485	439
	(b)	420	407	393	386	381	379	364
	(c)	468	455	430	424	418	454	438
$\Delta z = 0.02$	(a)	427	413	425	439	459	513	509
	(b)	423	409	392	388	383	381	369
	(c)	475	459	427	427	422	458	449
$\Delta z = 0.05$	(a)	401	399	391	383	391	383	358
	(b)	401	399	387	377	375	369	346
	(c)	430	439	417	405	406	434	403
$\Delta z = 0.2$	(a)	374	362	361	357	360	318	302
	(b)	374	362	361	355	360	318	301
	(c)	377	366	365	362	377	332	313

Table 9: $\sigma_v^{\rm max}$ in km/s: the dispersion each group reaches, by the linear response of Eq. (25), when the paper fit absorbs a truth $\alpha_{2,i} = -A$ in every bin, with (a) $A = 0.5/k_{\rm max}^2\, h^{-2}\,{\rm Mpc}^2$, (b) $12.5\, h^{-2}\,{\rm Mpc}^2$ and (c) $25\, h^{-2}\,{\rm Mpc}^2$, maximized over the cuts $k_{\rm max} = 0.05$–$0.2\, h$/Mpc at $z = 0$. The first two rows give the fiducial dispersions and the widened (paper) prior widths of Section 2.

($N_{\rm eval} = 172$), and DSATUR finds the four. These are linear-response estimates extrapolated over 100–200 km/s – the blur is not linear in σ_v – and the wider question, what the lists at the prior edge measure, is answered by the forecasts of `notes/cross_correlations` (its σ_v axis).

The cost. A longer pair list is a new data vector, so a layout needs its pair list, mask, covariance and scale cuts rebuilt, and then a new base. Table 10 prices the bases, scaling each layout's $t_{\rm eval}$ of Section 6.4 with the number of gg spectra – an upper estimate, since the bin-count scaling used there would leave $t_{\rm eval}$ unchanged. Measuring every within-sample $C^{i,i+2}$ costs about 6.7 hours of bases with four batches, or 8.2 hours with the greedy coloring's six, against about 4.9 hours today. Truth (a) forces one rebuild, $\Delta z = 0.02$: about 2.4 hours with four batches, 2.6 with the greedy five, against 2.1 today.

The adopted coloring. The code keeps the greedy coloring in bin order: it is simple to explain and to check, and on today's lists it already reaches the minimum. On the extended lists with every within-sample $C^{i,i+2}$ measured – the first chain of the gap axis that `notes/cross_correlations` forecasts – it overshoots the four-batch minimum with six batches, 190 evaluations per base against 154, an accepted cost of 18 evaluations per extra batch. On the $\sigma_v = 600$ km/s list of $\Delta z = 0.02$, the greedy rule, DSATUR and the clique bound all give six batches, so nothing is lost there. A switch to DSATUR would have to change the notebook and `scripts/compute_fisher_base_mpi.py` together, since the two copies must stay identical for the MPI caches to reproduce the notebook's; DSATUR stays an open item for a future study (Appendix B).

layout	gg spectra	t_{eval} [s]	base [h] today	4 batches	greedy
$\Delta d_{\text{com}} = 62$ Mpc	$183 \to 253$	56.7	1.75	2.42	2.99
$\Delta z = 0.02$	$223 \to 307$	67.5	2.10	2.89	3.56
$\Delta z = 0.05$	$88 \to 118$	26.1	0.83	1.12	1.38
$\Delta z = 0.2$	$23 \to 27$	6.3	0.23	0.27	0.27
all four			4.9	6.7	8.2

Table 10: The base of each layout with every within-sample $C^{i,i+2}$ measured: the number of gg spectra (today $\to$ extended), the estimated time per evaluation (t_{eval} of Section 6.4 scaled with the number of spectra, on the same Apple M2 Pro yardstick), and the base in hours – today's, with the four-batch coloring, and with the greedy coloring's batches (six, except four on $\Delta z = 0.2$).

7 A scale cut is a set of rows

A scale cut removes the band powers that small, nonlinear scales dominate. A cut at $k_{\max}$, quoted at $z = 0$, cuts bin i at $k_{\max}/D(\bar{z}_i)$, by the non-Limber ultraviolet-fraction criterion of Saraivanov et al. [3] within the shipped mask: a band is kept while less than 5% of it comes from the power spectrum above that wavenumber. The Fisher matrix at a cut follows in four steps.

1. The cut keeps a set K of the rows of m. It drops the other rows of B and keeps every column: a cut never changes a derivative. B_K is the kept rows of B, and C_{KK} the kept block of the covariance.

2. The data part of the Fisher matrix of a model at this cut is $F_{\text{data}} = \mathsf{D}_K^{\mathsf{T}} f_A \mathsf{C}_{KK}^{-1} \mathsf{D}_K$, with $f_A = 12{,}300/14{,}000$ the factor that rescales the inverse covariance to the survey area (Section 3). Since $\mathsf{D}_K = \mathsf{B}_K \mathsf{M}$,

$$F_{\text{data}} \;=\; \mathsf{M}^{\mathsf{T}} \big(f_A \, \mathsf{B}_K^{\mathsf{T}} \, \mathsf{C}_{KK}^{-1} \, \mathsf{B}_K \big) \mathsf{M} \;=\; \mathsf{M}^{\mathsf{T}} \, \mathsf{G}_K \, \mathsf{M}, \qquad \mathsf{G}_K \;\equiv\; f_A \, \mathsf{B}_K^{\mathsf{T}} \, \mathsf{C}_{KK}^{-1} \, \mathsf{B}_K. \tag{23}$$

   G_K is what the cut's kept data say about every pair of knobs: its entry (a, b) is the information shared by knobs a and b.

3. G_K contains neither M nor a prior. It depends only on the base, the cut and the covariance, so the notebook computes it once per cut, and every model and prior at that cut reuses it.

4. The priors add their precision: $F = \mathsf{M}^{\mathsf{T}} \mathsf{G}_K \mathsf{M} + F_{\text{prior}}$.

The toy, cut. Suppose the cut drops the cross spectrum C^{23}, which fails the ultraviolet criterion. Then:

- $K = \{C^{11}, C^{22}, C^{33}, C^{12}\}$, and B_K is the 4×7 matrix of the first four rows of Eq. (7);
- C_{KK} is 4×4, and G_K is 7×7;
- the group model's F is $(6 \times 7)(7 \times 7)(7 \times 6) = 6 \times 6$, and the free model's 7×7, both from the same G_K.

After this cut, v_3 survives only on C^{33}.

The real sizes. On $\Delta d_{\rm com} = 62$ Mpc, B is $10{,}494 \times 547$. The shipped mask keeps 5,674 rows, so B_K is $5{,}674 \times 547$, C_{KK} is $5{,}674 \times 5{,}674$ and G_K is 547×547. F is 138×138 for the paper model. The large block is handled once per cut, and every Fisher system at that cut (one Fisher matrix with its inverse and results, for each model and prior the notebook evaluates) reuses G_K.

How G_K is computed. The notebook computes G_K in `layout_grams`, without ever inverting the covariance:

1. it writes $\mathsf{C}_{KK} = LL^{\mathsf{T}}$ with L lower triangular, the Cholesky factorization, a matrix analogue of the square root;
2. it solves $L\,W = \mathsf{B}_K$ for W row by row, which a triangular L allows (forward substitution);
3. it forms $\mathsf{G}_K = f_A\, W^{\mathsf{T}} W$, with f_A the area rescaling of the inverse covariance.

This saves work, avoids amplifying round-off when the covariance is ill-conditioned (when its inverse magnifies small errors), and makes G_K symmetric with no negative information, by construction.

Each layout gets eight such matrices: the seven cuts $k_{\rm max} = 0.05, 0.075, \ldots, 0.2\,h/\mathrm{Mpc}$, and the shipped mask alone, the comparison point of the information test below.

The priors. $F_{\rm prior}$ adds precision and needs no data. The diagonal Gaussian priors, all centered on zero, sit on σ_v and on the model's group parameters (Section 2): the σ_v widths are the exploration fits' for the α_2-free variants, and six times them for the paper model, the paper yaml's own widths. The counterterm widths, $0.5/k_{\rm max}^2$, change with the cut, so the diagonal is rebuilt for each cut (`prior_diagonal`).

The linear-bias prior acts on the block of the β_i. The baseline and pessimistic priors add, group by group, the penalty of their memory model on bin-to-bin wiggles of the linear biases (`group_bias_precision`; the correlated-prior note, `notes/correlated_bias_prior_v2`, on what the prior charges and how it enters the Fisher matrix). The free prior adds nothing, and the growth-tied prior is a change of M.

Parameters without data. At small cuts some bins keep no band, and their columns vanish on the kept rows K. A parameter whose information at the cut, data plus prior, falls below 10^{-10} of its information on the shipped mask with the same priors leaves the fit: it is removed from M and from F, and the quoted FoMs, errors and shifts are over the remaining parameters. A Gaussian prior keeps a parameter in. A cut that keeps no band (as $\Delta z = 0.02$ at $k_{\rm max} = 0.05\,h/\mathrm{Mpc}$) or leaves a cosmological parameter without data gives no fit.

The pseudo-inverse. A plain inverse fails in one more case. On $\Delta d_{\rm com} = 62$ Mpc at $k_{\rm max} = 0.075\,h/\mathrm{Mpc}$ only 7 of the 18 BGS bins keep a band. A flat-prior BGS tilt can then lose every band that carries information on it, leaving one combination of parameters with no information at all. Along such a direction a plain inverse amplifies round-off by about 10^{16} and can turn it into a spurious shift of sigmas.

`stable_inverse` therefore takes a pseudo-inverse, an inverse that skips the directions with no information:

1. it scales F to unit diagonal, $\tilde{F} = SFS$ with $S = \mathrm{diag}(F)^{-1/2}$, so that the parameters' units do not matter;

2. it diagonalizes $\tilde{F} = V\Lambda V^{\mathsf{T}}$: the columns of V are the parameter combinations measured independently, and the eigenvalues in Λ the information in each;

3. it drops the directions whose eigenvalue is below 10^{-12} of the largest, and returns

$$F^{-1} \;=\; S\, V_\kappa\, \Lambda_\kappa^{-1}\, V_\kappa^{\mathsf{T}}\, S, \tag{24}$$

   with κ the kept directions.

A dropped direction must leave (w_0, w_a), Ω_m and A_s alone: the scan gives no fit when one touches them by more than 10^{-6}. The case above, the flat BGS tilts on $\Delta d_{\mathrm{com}} = 62$ Mpc at $k_{\max} = 0.075\,h/\mathrm{Mpc}$, is the mechanism. Across the notebook's $k_{\max}$ scan, 11 of its 168 Fisher systems drop a direction, each touching the scored parameters by at most 3.3×10^{-10}. The untied comparison of the summary note drops more. Its version with correlated residuals loses the residual trends that duplicate the flat group tilts: 18 directions at most cuts on the three thinner layouts, and up to 33 at the smallest cuts, each touching the scored parameters by at most 1.1×10^{-7}.

8 Everything downstream reuses G

Once G_K is built, nothing in the notebook touches an $n \times n$ matrix or the data vector again. F is Eq. (23) plus the priors, inverted by Eq. (24), and σ_j and the FoMs follow as in Section 3.

Shifts. A truth is an assumed true universe that departs from the fitted model, for example one with the redshift-space counterterm $\alpha_{2,i} \neq 0$ when the fit sets $\alpha_{2,i} = 0$ (Section 4). It moves the data by δm. When δm lies in the span of the base columns, $\delta m = \mathsf{B}c$, with c a vector of coefficients:

- $\alpha_{2,i} = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every bin i multiplies the exact per-bin α_2 columns of the base (Section 6.3) by 12.5;
- an offset truth puts its values on the $b_{2,i}$ and $b_{s^2,i}$ blocks;
- a fractional drift δ_i of the linear biases is $c = \sum_i \beta_i\,\delta_i\,\mathsf{M}_{\beta_i}$, with M_{β_i} the column of Eq. (13).

To first order the best fit then moves by

$$\delta p \;=\; F^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{C}^{-1}\delta m \;=\; F^{-1}\mathsf{M}^{\mathsf{T}}\,\mathsf{G}_K\,c \tag{25}$$

(the correlated-prior note, on the shift formula). $\delta m = \mathsf{B}c$ is exact for the $\alpha_{2,i}$, $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ truths, whose terms are linear, and first order for truths in $b_{1,i}$, $b_{2,i}$ and $b_{s^2,i}$, since the band powers are quadratic in them. A truth costs one product with G_K.

Random truths. Eq. (25) is linear in c, so a family of random truths collapses into one matrix: for drifts δ_i of the linear biases the (w_0, w_a) shift is $\delta w = A\,\delta$, with A the (w_0, w_a) rows of Eq. (25) applied to the N columns $\beta_i\mathsf{M}_{\beta_i}$. Two thousand truths cost one matrix product.

The exact $\langle\chi^2\rangle$. Whether a fit's error bars are honest is tested by averaging $\chi^2_w = (\hat{w} - w_{\rm true})^{\mathsf{T}}\mathsf{C}_w^{-1}(\hat{w} - w_{\rm true})$, with $\hat{w}$ the fitted (w_0, w_a), over the truths and the data noise. Every source is Gaussian and enters linearly, so the average is exact in closed form (the correlated-prior note, on the test statistic):

$$\langle\chi^2_w\rangle \;=\; \Delta^2_{\rm drift} \;+\; \mathrm{tr}\Big[\mathsf{C}_w^{-1}\left(A\,\Sigma\,A^{\mathsf{T}} + R\,\Sigma_\theta\,R^{\mathsf{T}} + N_w\right)\Big]\,. \tag{26}$$

- $\Delta_{\rm drift}$ is the (w_0, w_a) shift, in units of the ellipse, of the known drift alone: the smooth change of the linear biases with redshift that physics predicts.
- Σ is the covariance of the truth's remaining scatter of the linear biases.
- R and Σ_θ are the response to, and the prior variance of, any other parameter drawn at random in the accuracy test.
- N_w is the (w_0, w_a) block of $F^{-1}F_0F^{-1}$, the share of the data noise and the Gaussian priors, with F_0 the Fisher matrix without the linear-bias prior.

The notebook reports $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$, with $n_{\rm dof} = 2$ for (w_0, w_a), and $n_{\rm dof} = 1$ for Ω_m or A_s with that parameter's own row; 1 means honest error bars. The random draws only make the percentile bars of the correlated-prior note's figures.

The cost. Each G_K costs one Cholesky factorization of at most 7,621 rows, about $n_K^3/3 \approx 1.5\times10^{11}$ floating-point operations, and one triangular solve, eight times per layout. A Fisher system then costs $\mathsf{M}^{\mathsf{T}}\mathsf{G}_K\mathsf{M}$, with $n_{\rm base} \le 645$ and $n_{\rm par} \le 164$ (the fiducial model with α_2 free on $\Delta z = 0.02$), about $2\,n_{\rm base}^2 n_{\rm par} \approx 1.4\times10^8$ operations, plus the diagonalization of an $n_{\rm par}\times n_{\rm par}$ matrix: a fraction of a second.

The $k_{\rm max}$ scan of the notebook has 168 systems (four layouts, seven cuts, the fiducial model with α_2 free, with and without the tilts, and the baseline, pessimistic and free priors; the growth-tied prior fails the accuracy test and is left out), 162 with a fit and 6 without any band. In operations, all of them together cost less than the Cholesky factorizations of one layout. Without the base, each variant would need its own derivatives, at the cost of Section 6.4 per layout, and each truth its own data vector.

8.1 What is precomputed, and what is computed at every run

The work splits into three levels.

1. **On disk, once per layout (hours):** the base columns and the fiducial vector $m(p_0)$. Table 11 lists every array on $\Delta d_{\rm com} = 62$ Mpc.
2. **In memory, at every run (seconds):** for each scale cut, its kept rows K and the matrix $\mathsf{G}_K = f_A\,\mathsf{B}_K^{\mathsf{T}}\mathsf{C}_{KK}^{-1}\mathsf{B}_K$, with $f_A = 12{,}300/14{,}000$ the factor that rescales the inverse covariance to the survey area (Section 3), 547×547 on $\Delta d_{\rm com} = 62$ Mpc (Section 7). The covariance is read fresh from its file at every run.
3. **For every model and prior (a fraction of a second):** M, then $F = \mathsf{M}^{\mathsf{T}}\mathsf{G}_K\mathsf{M} + F_{\rm prior}$, its inverse, the FoMs, the shifts and $\langle\chi^2\rangle$ (Section 7 and this section). No evaluation of the data vector at all.

array	shape	what it is
`model_derivatives_dcom62_final.npz` (0.97 MB)		
`m0`	10,494	the fiducial band powers $m(p_0)$
`D`	$10{,}494 \times 20$	stencil columns: 7 cosmology, 6 B_g (check only), 7 σ_v
`ops['b1']`	$10{,}494 \times 76$	column i: $\partial m/\partial b_{1,i}$, the linear bias of bin i alone
`ops['b2']`	$10{,}494 \times 76$	column i: $\partial m/\partial b_{2,i}$
`ops['bs2']`	$10{,}494 \times 76$	column i: $\partial m/\partial b_{s^2,i}$
`ops['a0']`	$10{,}494 \times 76$	column i: $\partial m/\partial \alpha_{0,i}$, per $h^{-2}\,\mathrm{Mpc}^2$
`T`	$10{,}494 \times 76$	column i: $\partial m/\partial \mathrm{SN}_{0,i}$
`b1`, `b2`, `bs2`	76 each	the fiducial $b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$ of every bin
`names`, `fiducial`	20 each	the stencil parameters and their fiducial values
`priors`, `leaders`		the σ_v prior widths; each bin's group amplitude
`model_templates_dcom62_final.npz` (0.22 MB)		
`m0`	10,494	the fiducial vector again (the cross-cache check)
`a2`	$10{,}494 \times 76$	column i: $\partial m/\partial \alpha_{2,i}$, per $h^{-2}\,\mathrm{Mpc}^2$, exact
`T0`	$10{,}494 \times 76$	the $\mathrm{SN}_{0,i}$ templates again (equal to `T`)
`T2`	$10{,}494 \times 76$	column i: $\partial m/\partial \mathrm{SN}_{2,i}$
`nonlinearity`, `reset`		the α_2 forward-backward check; $\max\lvert m_{\mathrm{after}} - m_0\rvert / \max\lvert m_0\rvert$

Table 11: The precomputed arrays of $\Delta d_{\mathrm{com}} = 62$ Mpc, in compressed `.npz` files of plain numpy arrays (Section 11): $n = 10{,}494$ band powers and $N = 76$ bins. Each $10{,}494 \times 76$ block is 6.4 MB in double precision in memory; the files are far smaller because no per-bin column has a nonzero entry outside its own bin's spectra. The files also hold the settings, tier and fingerprint they were made with (Section 9). The other layouts' files are 1.28/0.29, 0.49/0.11 and 0.13/0.03 MB ($\Delta z = 0.02$, 0.05, 0.2; derivatives/templates), and the scale-cut working copy is 3.6 MB. In the toy's notation, `ops['b1']` is $[u_1, \ldots, u_N]$ and `ops['b2']` is $[v_1, \ldots, v_N]$.

What forces the saved derivatives to be recomputed. Only what changes the knobs' responses: the fiducial point, the likelihood settings (accuracy, options, parameter files) and the code. The fingerprint of Section 9 catches changes of the input; a change of the code needs a flag set by hand (Section 11). The recipe M, the priors, the scale cut and the covariance file never force a recompute.

9 Derivative accuracy

Why a derivative needs more than a χ^2. Every evaluation of m carries numerical noise: the integration grids, the non-Limber distance grid and CAMB's tolerances change the vector by small amounts that do not vary smoothly with the parameters. A finite difference divides that noise by the step, so a setting that converges a χ^2 (the MCMC criterion is $|\Delta\chi^2| < 0.2$) can leave a derivative noisy.

The noise biases the forecast one way. A column $d_j + e_j$, with e_j its noise, gives $F_{jj} = (d_j + e_j)^{\mathsf{T}}\mathsf{C}^{-1}(d_j + e_j)$, whose average exceeds the true value by $\langle e_j^{\mathsf{T}}\mathsf{C}^{-1}e_j\rangle > 0$: noise looks like information, and the FoM comes out too high.

The study: a starting configuration, and one control at a time. The convergence study, `tests/studies/run_fisher_derivatives.py`, measures the effect on $\Delta z = 0.05$. It scores the marginalized errors of w_0, w_a, Ω_m and A_s, both FoMs, and the shifts under truths

that carry a term the fit lacks or holds with a prior (the models are defined in the higher-order note, `notes/higher_order_bias`), at $k_{\max} = 0.075$ and $0.2\,h/\mathrm{Mpc}$. The FoM offsets of the grid scan, the step test, the final settings and DerivKit are the paper model's, scored from the study's derivatives on the paper-fiducial cuts (the `data.json` behind Figure 5). The error and shift stabilities, and the offsets of the other controls (CAMB's boost among them), are scored on the study's own system, the fiducial model with α_2 free and the exploration σ_v widths; the text marks each such number. The derivative columns do not depend on the model, so both scorings test the same columns (Appendix B).

The scan starts from one configuration, the starting configuration: a grid factor $g = 128$ with every other control raised, CosmoLike's `accuracyboost` 4 and `integration_accuracy` 2, the low-rank engine's rank 8 (the number of terms kept in its sum) and multipole cap 2000, and CAMB's `AccuracyBoost` 2.0, `k_per_logint` 30 and maximum wavenumber $80\,\mathrm{Mpc}^{-1}$. The scan then varied one control at a time away from it. Here g is the product `accuracyboost` $\times$ `nonlimber_accuracyboost`, and the non-Limber projection integrates over about $511\,g$ comoving distances; the MCMC value on $\Delta z = 0.05$ is $g = 32$. Each control was lowered toward its MCMC value; the grid was also raised to 256 and 512. The five-point stencil at step h on $g = 512$ is the converged value every result is compared with. The final tier, chosen at the end of the section, uses $g = 256$. (The study's result files name the two endpoints by key: `reference` is the starting configuration and `candidate` the final settings.)

The noise of one evaluation. The computed vector is not the smooth model alone. Write

$$m_{\mathrm{c}}(\theta) \;=\; m(\theta) \;+\; n(\theta), \tag{27}$$

with m the model, smooth in the parameters, and n the numerical error of one evaluation, from the causes named in "Why a derivative needs more than a χ^2": the integration grids, the non-Limber distance grid and CAMB's tolerances. n is small, but it does not vary smoothly with θ: a tiny move of the parameters shifts where the model falls on the fixed grids, and the error changes irregularly from point to point. The study measures the size of n from the five stencil points themselves (`stencil_noise` in `tests/studies/run_fisher_derivatives.py`), with no extra evaluations. The next paragraphs build the estimator one step at a time.

The fourth difference removes the smooth part. Take the five equally spaced points $\theta_0 + k\epsilon$, $k = -2, \ldots, 2$, and combine them with the weights $c_k = (1, -4, 6, -4, 1)$, the binomial coefficients of $(1 - x)^4$ with alternating signs:

$$\Delta^4 m \;=\; \sum_{k=-2}^{2} c_k\, m(k) \;=\; m(-2) - 4\,m(-1) + 6\,m(0) - 4\,m(1) + m(2), \tag{28}$$

writing $m(k)$ for $m_{\mathrm{c}}(\theta_0 + k\epsilon)$. Expand each point of the smooth part in a Taylor series around θ_0,

$$m(\theta_0 + k\epsilon) \;=\; \sum_{n=0}^{\infty} \frac{(k\epsilon)^n}{n!}\, m^{(n)}(\theta_0), \tag{29}$$

and collect the terms order by order: the n-th derivative enters $\Delta^4 m$ with the coefficient $(\epsilon^n/n!)\sum_k c_k\, k^n$. These sums can be checked by hand:

n	0	1	2	3	4	5	6
$\sum_k c_k\, k^n$	0	0	0	0	24	0	120

For $n = 4$ the sum is $16 - 4 + 0 - 4 + 16 = 24$; every odd order vanishes because the weights are symmetric, $c_{-k} = c_k$, while k^n is odd. So everything up to the third derivative cancels exactly, the fourth order survives with $\epsilon^4\,(24/4!) = \epsilon^4$, and the next surviving order is $n = 6$ with $120/6! = 1/6$:

$$\Delta^4 m\big|_{\text{smooth}} \;=\; \epsilon^4\, m''''(\theta_0) \;+\; \frac{\epsilon^6}{6}\, m^{(6)}(\theta_0) \;+\; \ldots \tag{30}$$

The study's steps are a few percent of each parameter, and the model varies smoothly on the scale of the parameter itself, so $\epsilon^4 m''''$ is suppressed by ϵ^4, roughly seven orders of magnitude: for a smooth vector the fourth difference is essentially zero. What survives in Eq. (28) is the noise.

Independent noise adds in quadrature: the 70. The noise part of the fourth difference is $\sum_k c_k\, n_k$, with n_k the error at point k. Errors that change irregularly from point to point are modelled as independent draws with a common variance σ^2. The variance of a weighted sum of independent terms is the sum of the squared weights times σ^2:

$$\mathrm{Var}\Big(\sum_k c_k\, n_k\Big) \;=\; \sigma^2 \sum_k c_k^2 \;=\; (1 + 16 + 36 + 16 + 1)\,\sigma^2 \;=\; 70\,\sigma^2. \tag{31}$$

So $|\Delta^4 m|/\sqrt{70}$ estimates σ, the noise of one evaluation. One stencil is enough: the data vector has thousands of entries, the fourth difference is taken entry by entry, and the length of the result averages over all of them, so a single realization gives a tight estimate.

A toy, with invented numbers. Take one entry, the smooth model $m(\theta) = e^\theta$ at $\theta_0 = 0$ with $\epsilon = 0.01$, and plant noise of true rms 0.8×10^{-6} at the five points: $n_k = (1.3, -0.4, 0.9, -0.6, 0.2) \times 10^{-6}$. The smooth part of the fourth difference is $\epsilon^4\, e^{\theta_0} = 1.0 \times 10^{-8}$, a thousand times below the noise combination. The noise part is $1.3 + (-4)(-0.4) + 6\,(0.9) + (-4)(-0.6) + 0.2 = 10.9$, times 10^{-6}, so the estimate is $10.9 \times 10^{-6}/\sqrt{70} = 1.3 \times 10^{-6}$: the right size, though off the true 0.8×10^{-6} by a factor 1.7, as a single entry must be, since one number is estimating a variance. Averaged over the thousands of entries of a real data vector, that scatter disappears.

Measuring a vector in the data's own units. A raw $|\Delta^4 m|$ would mix entries with very different error bars, and neighbouring band powers are correlated. The cure is to transform the vector first, so that, measured against the data's covariance C, the error of every entry is one and no two entries are correlated. After the transformation each entry counts in units of its own error bar, and the plain sum of squares of the transformed vector is a χ^2.

The transformation, through the eigenvectors. Diagonalize the covariance, $\mathsf{C} = U \Lambda U^{\mathsf{T}}$, with U the orthogonal matrix of its eigenvectors and Λ the diagonal matrix of the variances along them. For a vector v, the rotation $U^{\mathsf{T}} v$ gives its components along the eigenvectors, the

directions in which the data's errors are uncorrelated, and $\Lambda^{-1/2}$ then divides each component by its standard deviation. So

$$y \;=\; \Lambda^{-1/2}\,U^{\mathsf{T}}\,v \tag{32}$$

has unit, uncorrelated errors, and $|y|^2 = v^{\mathsf{T}}U\Lambda^{-1}U^{\mathsf{T}}v = v^{\mathsf{T}}\mathsf{C}^{-1}v$. This transformation is called *whitening*, after white noise: noise with equal power in every direction, as white light holds every colour.

Why a Cholesky factor does the same job. The code whitens with the Cholesky factorization instead, $\mathsf{C} = LL^{\mathsf{T}}$ with L lower triangular (a matrix analogue of the square root): $y = L^{-1}v$ has covariance $L^{-1}\,\mathsf{C}\,L^{-\mathsf{T}} = 1$, so its errors are also unit and uncorrelated, and its length is the same, $|L^{-1}v|^2 = v^{\mathsf{T}}\mathsf{C}^{-1}v$. The two transforms differ only by a rotation, which changes no length; the Cholesky factorization is cheaper and numerically stable, so the code uses it.

A 2×2 toy. Two band powers with error bars 1 and 2 and correlation 0.5 have

$$\mathsf{C} = \begin{pmatrix} 1 & 1 \\ 1 & 4 \end{pmatrix}, \qquad L = \begin{pmatrix} 1 & 0 \\ 1 & \sqrt{3} \end{pmatrix}, \qquad L^{-1} = \begin{pmatrix} 1 & 0 \\ -1/\sqrt{3} & 1/\sqrt{3} \end{pmatrix} \tag{33}$$

(the off-diagonal entry is $0.5\times 1\times 2 = 1$; multiplying out LL^{T} recovers C). Take the difference vector $v = (1, 1)$: whitened, $y = L^{-1}v = (1, 0)$, and $|y|^2 = 1$. Directly, $\mathsf{C}^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 4 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ and $v^{\mathsf{T}}\mathsf{C}^{-1}v = (4 - 1 - 1 + 1)/3 = 1$: the same χ^2. Counting each entry against its own error bar alone would give $1/1 + 1/4 = 1.25$; the two entries are correlated, so part of the misfit repeats and whitening counts it once. The eigenvector route, with variances $(5 \pm \sqrt{13})/2 \approx 4.30$ and 0.70, gives a different y of the same length 1: the two transforms differ by a rotation.

The estimate in these units. The study whitens the fourth difference with the shipped-mask covariance at the survey area (f_A applied) and reports

$$\hat{\sigma} \;=\; \left|L^{-1}\,\Delta^4 m\right| / \sqrt{70}\,: \tag{34}$$

the numerical error of one evaluation, measured in the data's own error bars ($\sqrt{\chi^2}$ units), listed per stencil parameter in Table 12. Reading one entry: 1.4×10^{-2} for w_a at the MCMC settings means that the numerical error of one evaluation along the w_a stencil moves the whitened vector by 0.014, a $\Delta\chi^2$ of 2×10^{-4} – far below the MCMC convergence criterion $|\Delta\chi^2| < 0.2$, and harmless for a likelihood evaluation. The next paragraph is why it still matters.

Why the jitter matters for a derivative. First the objects. The Fisher matrix, Eq. (2), is built from the derivative matrix D, one column per sampled parameter p_j. The column the code wants is $d_j = \partial m/\partial p_j$ at the fiducial point: the exact derivative of the smooth model. The column it gets applies the stencil, Eq. (14), to the computed vectors of Eq. (27), $m_{\mathrm{c}} = m + n$. The smooth parts combine to d_j (up to the ϵ^4 truncation of the stencil), the noise parts to an error, so the computed column is $d_j + e_j$ with

$$e_j \;=\; \frac{n(-2) - 8\,n(-1) + 8\,n(1) - n(2)}{12\,\epsilon_j}, \tag{35}$$

configuration	$10^9 A_s$	H_0	Ω_m	w_0	w_a
MCMC settings	6.0×10^{-6}	2.1×10^{-3}	1.7×10^{-2}	2.5×10^{-2}	1.4×10^{-2}
$g = 64$	4.5×10^{-6}	2.2×10^{-3}	8.1×10^{-3}	1.1×10^{-2}	4.8×10^{-3}
$g = 128$ (starting)	4.5×10^{-6}	2.2×10^{-3}	6.5×10^{-3}	4.3×10^{-3}	1.8×10^{-3}
$g = 256$	4.5×10^{-6}	2.2×10^{-3}	6.1×10^{-3}	2.2×10^{-3}	3.5×10^{-4}
final settings	4.6×10^{-6}	3.4×10^{-3}	3.1×10^{-3}	5.8×10^{-4}	3.2×10^{-4}

Table 12: The noise of one evaluation along each stencil parameter, in $\sqrt{\chi^2}$ units, on $\Delta z = 0.05$. The rows with a grid factor g keep every other control at the starting configuration's value ($g = 128$ is the starting configuration itself); the final settings change the other controls as well, at the final grid factor $g = 256$. The five columns shown are the stencil parameters the FoM pairs read most; the other nine (n_s, Ω_b, the seven σ_v) are measured alongside and enter the prediction of Table 16. The unit is the natural one for a Fisher forecast: the data vector enters $F = \mathsf{D}^\mathsf{T}\mathsf{C}^{-1}\mathsf{D}$ only through the metric C^{-1}, so a change δm matters only through its whitened length $\sqrt{\delta m^\mathsf{T}\mathsf{C}^{-1}\delta m}$, and a column whose noise is σ per evaluation adds about $0.9\,(\sigma/\epsilon)^2$ to F_{jj}, Eq. (37).

writing $n(k)$ for the evaluation noise k steps from the fiducial along p_j, and ϵ_j for that parameter's step. Like d_j, the error e_j is a vector with one entry per band power. And each column has its own: the four evaluations behind the column of another parameter p_k move only p_k (the center point carries weight zero and is never evaluated), so e_k is built from its own evaluations, independent of those behind e_j.

What the noise does to F. Put the computed column into Eq. (2):

$$F_{jj} \;=\; (d_j + e_j)^\mathsf{T}\mathsf{C}^{-1}(d_j + e_j) \;=\; d_j^\mathsf{T}\mathsf{C}^{-1}d_j \;+\; 2\,d_j^\mathsf{T}\mathsf{C}^{-1}e_j \;+\; e_j^\mathsf{T}\mathsf{C}^{-1}e_j. \tag{36}$$

The first term is the true entry. The middle term is linear in the noise, and the noise scatters with no preferred sign (the independent draws of "Independent noise adds in quadrature: the 70"), so it averages to zero. The last term is a sum of squares, like a χ^2: positive for every noise realization, so on average the noise only *adds* to F_{jj} – the one-way bias of "Why a derivative needs more than a χ^2": noise looks like information. An off-diagonal entry F_{jk} gains $e_j^\mathsf{T}\mathsf{C}^{-1}e_k$ and two linear cross terms; e_j and e_k come from separate evaluations, so all three average to zero. The systematic push sits on the diagonal.

How large the push is. The four errors in the numerator of Eq. (35) are independent, so their whitened lengths add in quadrature through the squared weights, $1 + 64 + 64 + 1 = 130$, and the whitened length of e_j is about

$$\frac{\sqrt{1+64+64+1}}{12}\,\frac{\sigma}{\epsilon_j} \;=\; \frac{\sqrt{130}}{12}\,\frac{\sigma}{\epsilon_j} \;\approx\; 0.95\,\frac{\sigma}{\epsilon_j}, \tag{37}$$

with σ the whitened ($\sqrt{\chi^2}$-unit) noise of one evaluation along that column's stencil, Table 12: a derivative divides the jitter by its step. Squaring gives the average of the spurious term of Eq. (36), $\langle e_j^\mathsf{T}\mathsf{C}^{-1}e_j\rangle \approx 0.9\,(\sigma/\epsilon_j)^2$; with σ in $\sqrt{\chi^2}$ units and ϵ_j in the parameter's units, the addition is in one over the parameter's units squared, the units of F_{jj}.

In numbers, along w_a. Its fiducial value is zero and its prior flat, so its step is absolute and unscaled, $\epsilon_{w_a} = h = 0.02$ ("The cosmological and σ_v columns"). At the MCMC settings the w_a noise is $\sigma = 1.4 \times 10^{-2}$, Table 12, so the column adds a spurious $0.9\,(0.014/0.02)^2 \approx 0.44$ to

	w_a		$10^9 A_s$	
	$k_{\max} = 0.075$	0.2	0.075	0.2
F_{jj} (others fixed)	670	2000	2.5×10^4	5.9×10^4
$1/\sigma_j^2$ (marginalized)	0.013	0.18	0.048	0.87
a_j, MCMC settings	0.41		1.8×10^{-8}	
a_j, final settings	2.4×10^{-4}		1.1×10^{-8}	
$a_j\sigma_j^2$, MCMC settings	33	2.3	3.8×10^{-7}	2.1×10^{-8}
$a_j\sigma_j^2$, final settings	0.019	1.3×10^{-3}	2.3×10^{-7}	1.3×10^{-8}

Table 13: The spurious additions against the real information on w_a and $10^9 A_s$, on the converged $\Delta z = 0.05$ paper-model system. F_{jj}: the information with every other parameter fixed. $1/\sigma_j^2$: the marginalized information. $a_j = 0.9\,(\sigma/\epsilon_j)^2$: the spurious addition, computed from the unrounded noise behind Table 12; it does not depend on the cut. $a_j\sigma_j^2$: the addition over the marginalized information, the dimensionless size. $k_{\max}$ in h/Mpc, quoted at $z = 0$.

$F_{w_a w_a}$ (w_a is dimensionless, so the entry is a plain number). At the final grid, $\sigma = 3.5 \times 10^{-4}$, the addition is $0.9\,(3.5 \times 10^{-4}/0.02)^2 \approx 2.8 \times 10^{-4}$. The spurious information grows as the noise *squared*: the fortyfold fall of the w_a noise in Table 12, from the MCMC settings to the final grid, is a 1600-fold fall of its spurious contribution to F.

And along $10^9 A_s$, the quietest column. Its noise is 4.6×10^{-6} at the final settings on a step $\epsilon = 0.02 \times 2.1 = 0.042$, so the addition is $0.9\,(4.6 \times 10^{-6}/0.042)^2 \approx 1 \times 10^{-8}$ – negligible. FoM$(\Omega_m, 10^9 A_s)$ therefore picks up evaluation noise through the Ω_m column and the other columns correlated with the pair, not through A_s itself.

Big or small? An addition needs a base, and Table 13 puts the two columns' additions next to the information the data actually carry, on the converged $\Delta z = 0.05$ paper-model system. Two bases offer themselves and they differ by factors of 10^4 to 10^5: the diagonal entry F_{jj}, the information on p_j with every other parameter held fixed, and the marginalized information $1/\sigma_j^2$, with σ_j the forecast error when everything else is free. The right yardstick is the marginalized one: the degeneracies make $1/\sigma_j^2$ far smaller than F_{jj}, and the fake prior competes with what is left after marginalization, not with the fixed-parameter information. By that yardstick the MCMC-settings addition is about 30 times the real marginalized information on w_a at $k_{\max} = 0.075\,h$/Mpc – the size the MCMC row of the measured-against-predicted comparison below registers – while the final-settings addition is about 2% of it, and the $10^9 A_s$ additions never exceed a few times 10^{-7} of theirs. The subsection below turns exactly this ratio, $a_j\sigma_j^2 = (\sigma_j/w_j)^2$, into the measure of the fake prior. A rule of thumb for what follows: once $a_j\sigma_j^2$ is below about 10^{-2} for every column, the noise moves the FoM by less than half a percent – that setting has settled.

Choosing the settings cheaply: how the evaluation noise reaches the FoM. The problem this subsection solves. Finding the accuracy settings at which the FoM stops changing normally takes a full derivative base at every candidate setting: 154 evaluations (Table 6) for the twenty sampled parameters plus the per-bin columns – about 50 minutes on $\Delta z = 0.05$ at the final settings (some 20 s per evaluation) and 1.8 to 2.1 hours per base on the thin layouts, so a scan over candidate settings costs hours to days. This subsection builds a cheap estimate instead: one low-accuracy base, computed once, and then, per candidate setting, only the noise

of a few heavy columns – about 9 to 21 evaluations, minutes instead of the better part of an hour.

First the question. The paragraphs above pin down what the noise does to the Fisher matrix: the noise of stencil column j adds, on average, about $0.9\,(\sigma/\epsilon_j)^2$ to the diagonal entry F_{jj}, and nothing, on average, anywhere else. But the published numbers are not Fisher entries; they are marginalized errors and FoMs. What does the addition do to $\mathrm{FoM}(w_0, w_a)$?

What the FoM reads. Recall Section 3: F^{-1} is the forecast covariance of the parameters, C_w is its 2×2 block for the pair (w_0, w_a), and $\mathrm{FoM}(w_0, w_a) = [\det \mathsf{C}_w]^{-1/2}$ is the inverse area of the pair's error ellipse when every other parameter is free. So the additions land in F, while the FoM reads a block of F^{-1}.

Why one cannot just look at F_{ww}. A block of the inverse is not the inverse of the block: $(F^{-1})_{ww} \neq (F_{ww})^{-1}$ whenever the pair correlates with any other parameter, because inverting F mixes every parameter into every block – that mixing *is* marginalization. The additions must therefore be followed through the inverse of the full matrix. The route below, each step a piece of the tool: the addition rewritten as a prior (a_j, the one ingredient a setting changes); the marginalization as a block formula (where the setting-independent σ_j enter); a 3×3 determinant and a worked toy (the estimate in its two-column form); and the general case of every column noisy at once (the weights R_j^2, and with them the low cost).

The noise as a fake prior. The addition has a familiar form. Below Eq. (2): a Gaussian prior of width w adds $1/w^2$ to its parameter's diagonal entry of F, and nothing else. The noise of column j adds, on average,

$$a_j \;=\; \langle e_j^{\mathsf{T}} \mathsf{C}^{-1} e_j \rangle \;\approx\; 0.9\,(\sigma/\epsilon_j)^2 \tag{38}$$

to F_{jj}, and nothing, on average, anywhere else (the cross terms of "What the noise does to F" have zero mean). Matching the two statements: the noise of column j acts exactly like a Gaussian prior on p_j with $1/w_j^2 = a_j$ – a spurious prior of width $w_j = a_j^{-1/2} \approx \epsilon_j/(0.95\,\sigma)$, information the data do not carry. For the tool, a_j is the one ingredient that depends on the candidate setting, and it is cheap: four evaluations along column j measure it.

In numbers, along w_a. At the MCMC settings the addition computed above is $a_{w_a} \approx 0.44$, a spurious prior of width $1/\sqrt{0.44} \approx 1.5$ on w_a; at the final grid, $a_{w_a} \approx 2.8 \times 10^{-4}$, a width of 60. Whether a prior of width 1.5 matters depends on how well the data measure w_a: a forecast feels a prior through the ratio of the parameter's marginalized error σ_j to the prior's width. The determinant expansion below makes that precise.

Marginalizing over the other parameters. Why the tool needs this step: the noise reaches the FoM only through the marginalized block, so the estimate will be built from marginalized quantities – σ_j here, R_j later – which barely depend on the accuracy setting and can be computed once, on a cheap system.

The block split. Order the sampled parameters so the pair $w = (w_0, w_a)$ comes first and the rest – call them o, for "others" – follow, and write the Fisher matrix in blocks,

$$F \;=\; \begin{pmatrix} F_{ww} & F_{wo} \\ F_{ow} & F_{oo} \end{pmatrix} :$$

F_{ww} is the pair's 2×2 corner, F_{oo} the square block of the other parameters among themselves, and $F_{wo} = F_{ow}^{\mathsf{T}}$ (two rows, one column per other parameter) their cross information. The pair's

block of F^{-1}, the C_w of the FoM, is the inverse of

$$F_w \;=\; F_{ww} - F_{wo}\,F_{oo}^{-1}\,F_{ow}, \qquad \mathsf{C}_w = F_w^{-1}\,: \tag{39}$$

F_w is the 2×2 information left on the pair when every other parameter is free – the subtraction is what marginalization costs – and $\mathrm{FoM}(w_0, w_a) = [\det \mathsf{C}_w]^{-1/2} = (\det F_w)^{1/2}$.

Where the formula comes from. Eq. (39) is a standard linear-algebra result, the inverse of a partitioned matrix: split any invertible matrix into four blocks,

$$M \;=\; \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \quad\Longrightarrow\quad (M^{-1})_{11} \;=\; \left(M_{11} - M_{12}\,M_{22}^{-1}\,M_{21}\right)^{-1},$$

whenever M_{22} is invertible too. The combination $M_{11} - M_{12}\,M_{22}^{-1}\,M_{21}$ is called the Schur complement of M_{22} in M; block matrices and their inverses are collected in the Matrix Cookbook [5]. Here $M_{11} = F_{ww}$, $M_{12} = F_{wo}$, $M_{21} = F_{ow}$ (F is symmetric), $M_{22} = F_{oo}$, and $\mathsf{C}_w^{-1} = F_w$ is the Schur complement of F_{oo}.

The proof, in three steps on F itself. First, name the unknowns: the first block column of F^{-1} holds C_w on top and, below it, the cross block $(F^{-1})_{ow}$, the forecast covariance of the other parameters with the pair. Second, read $F\,F^{-1} = I$ block row by block row on that column (I the identity):

$$F_{ww}\,\mathsf{C}_w + F_{wo}\,(F^{-1})_{ow} \;=\; I, \qquad F_{ow}\,\mathsf{C}_w + F_{oo}\,(F^{-1})_{ow} \;=\; 0.$$

Third, solve: the second equation gives $(F^{-1})_{ow} = -F_{oo}^{-1}\,F_{ow}\,\mathsf{C}_w$, and substituting it into the first leaves $\left(F_{ww} - F_{wo}\,F_{oo}^{-1}\,F_{ow}\right)\mathsf{C}_w = I$, which is Eq. (39).

The statistics behind the algebra. F_{ww} alone is the information on (w_0, w_a) with every other parameter held fixed, and the subtracted $F_{wo}\,F_{oo}^{-1}\,F_{ow}$ is the information the other parameters absorb when they are left free, that is, marginalized. Marginalizing a Gaussian keeps a block of its covariance F^{-1} (Section 3); written in terms of F, that block is this Schur complement.

Why the additions pass through the marginalization whole. The claim, then its construction. Let only the w_0 and w_a columns be noisy. Their additions land on the diagonal of F_w unchanged – exactly, not only to first order. This is what keeps the estimate to one line: no new matrix algebra per setting. Three block steps show it, and a 3×3 example then shows it entry by entry.

Step one, the noisy matrix in blocks. On average the noise adds a_{w_0} and a_{w_a} to the two diagonal entries of the pair and nothing else (the off-diagonal additions average to zero, as shown above), so the noisy Fisher matrix is

$$F' \;=\; F + \mathsf{E}, \qquad \mathsf{E} \;=\; \begin{pmatrix} \mathsf{A} & 0 \\ 0 & 0 \end{pmatrix}, \qquad \mathsf{A} \;=\; \begin{pmatrix} a_{w_0} & 0 \\ 0 & a_{w_a} \end{pmatrix},$$

with zeros in every block the noise does not touch.

Step two, the blocks of F'. Adding E changes one block and leaves three alone:

$$F'_{ww} = F_{ww} + \mathsf{A}, \qquad F'_{wo} = F_{wo}, \qquad F'_{ow} = F_{ow}, \qquad F'_{oo} = F_{oo}.$$

Step three, into the marginalization. Put these blocks into Eq. (39):

$$F'_w \;=\; (F_{ww} + \mathsf{A}) - F_{wo}\,F_{oo}^{-1}\,F_{ow} \;=\; F_w + \mathsf{A}.$$

No approximation was made: F_{ww} is the one block that enters Eq. (39) linearly and whole, and the subtracted term is built from F_{wo} and F_{oo} alone, which the noise leaves untouched.

The same, entry by entry. Take three parameters (w_0, w_a, o), so $F_{oo} = f_{33}$ is a number, and write the marginalization out:

$$F = \begin{pmatrix} f_{11} & f_{12} & f_{13} \\ f_{12} & f_{22} & f_{23} \\ f_{13} & f_{23} & f_{33} \end{pmatrix} \quad \Longrightarrow \quad F_w = \begin{pmatrix} f_{11} - f_{13}^2/f_{33} & f_{12} - f_{13}f_{23}/f_{33} \\ f_{12} - f_{13}f_{23}/f_{33} & f_{22} - f_{23}^2/f_{33} \end{pmatrix}.$$

The noise changes only $f_{11} \to f_{11} + a_{w_0}$ and $f_{22} \to f_{22} + a_{w_a}$:

$$F' = \begin{pmatrix} f_{11} + a_{w_0} & f_{12} & f_{13} \\ f_{12} & f_{22} + a_{w_a} & f_{23} \\ f_{13} & f_{23} & f_{33} \end{pmatrix}$$

$$\Longrightarrow \quad F'_w = \begin{pmatrix} f_{11} + a_{w_0} - f_{13}^2/f_{33} & f_{12} - f_{13}f_{23}/f_{33} \\ f_{12} - f_{13}f_{23}/f_{33} & f_{22} + a_{w_a} - f_{23}^2/f_{33} \end{pmatrix}.$$

None of the three subtracted pieces – f_{13}^2/f_{33}, $f_{13}f_{23}/f_{33}$, f_{23}^2/f_{33} – contains f_{11} or f_{22}, so $F'_w = F_w + \mathsf{A}$: each addition rides through the marginalization into its own diagonal entry.

A worked toy, part one: the marginalization. The toy the rest of this subsection reuses. Three sampled parameters, the pair (w_0, w_a) and one other parameter o standing in for all the others, with the round-number Fisher matrix

$$F = \begin{pmatrix} 5 & 4 & 1 \\ 4 & 5 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

in that order (all three parameters treated as dimensionless, so every entry is a plain number; the matrix is positive definite, with eigenvalues 0.71, 2.6 and 9.7). The entries are chosen so the forecast resembles the real one: a strong w_0–w_a anticorrelation and a sizeable correlation with o, computed below.

The marginalization, with the entry-by-entry formula. Here $f_{33} = 3$, and the three subtracted pieces are $f_{13}^2/f_{33} = 1/3$, $f_{13}f_{23}/f_{33} = 2/3$ and $f_{23}^2/f_{33} = 4/3$:

$$F_w = \begin{pmatrix} 5 - 1/3 & 4 - 2/3 \\ 4 - 2/3 & 5 - 4/3 \end{pmatrix} = \begin{pmatrix} 14/3 & 10/3 \\ 10/3 & 11/3 \end{pmatrix} \approx \begin{pmatrix} 4.67 & 3.33 \\ 3.33 & 3.67 \end{pmatrix},$$

with $\det F_w = (14 \times 11 - 10 \times 10)/9 = 6$, so $\mathrm{FoM} = \sqrt{6} \approx 2.45$.

Fixed versus free. The same matrices show what the marginalization costs. With o held fixed, the pair's information is the corner block F_{ww} itself: $\det F_{ww} = 5 \times 5 - 4 \times 4 = 9$, the FoM would be $\sqrt{9} = 3.00$, and the errors $\sigma_{w_0} = \sigma_{w_a} = \sqrt{5/9} \approx 0.745$. With o free, $\det F_w = 6$: $\mathrm{FoM} = \sqrt{6} \approx 2.45$, and the marginalized errors grow to $\sigma_{w_0} \approx 0.782$ and $\sigma_{w_a} \approx 0.882$ (part two below inverts F_w in full). The difference is the subtracted matrix,

$$F_{wo}\, F_{oo}^{-1}\, F_{ow} = \frac{1}{3}\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix},$$

the information o absorbs when it is left free – and it is why the FoM cannot be read off F_{ww}.

The pass-through, with numbers. Add $a_{w_0} = 0.5$ and $a_{w_a} = 0.3$. The direct route – form $F + \mathsf{E}$, redo the Schur complement with $f_{11} = 5.5$ and $f_{22} = 5.3$ – gives

$$F'_w \;=\; \begin{pmatrix} 5.5 - 1/3 & 4 - 2/3 \\ 4 - 2/3 & 5.3 - 4/3 \end{pmatrix} \;\approx\; \begin{pmatrix} 5.17 & 3.33 \\ 3.33 & 3.97 \end{pmatrix} \;=\; F_w + \begin{pmatrix} 0.5 & 0 \\ 0 & 0.3 \end{pmatrix},$$

the additions sitting whole on the diagonal, as the blocks promised. How much the FoM moved takes the determinant, next.

The two-column result, step by step. This paragraph is the estimate in its two-column form: a setting-independent weight, σ_j^2, times the setting-dependent a_j.

The FoM as one determinant. Stay with the 3×3 matrix of the construction above: the pair's rows first, one other parameter o, entries $f_{11} \ldots f_{33}$. The block result has a determinant form too, $\det F = \det F_{oo} \det F_w$ [5]; with a single other parameter $\det F_{oo} = f_{33}$, so $\mathrm{FoM}^2 = \det F_w = \det F / f_{33}$ (in the toy, $\det F_w = 18/3 = 6$). The noise leaves f_{33} untouched, so that factor cancels in the ratio:

$$\frac{\mathrm{FoM}'^2}{\mathrm{FoM}^2} \;=\; \frac{\det F'_w}{\det F_w} \;=\; \frac{\det F'}{\det F}.$$

The determinant, term by term. A determinant is linear in each row separately, and the noise touches one entry of row 1 and one of row 2:

$$\begin{aligned} \det F' \;&=\; \det \begin{pmatrix} f_{11} + a_{w_0} & f_{12} & f_{13} \\ f_{12} & f_{22} + a_{w_a} & f_{23} \\ f_{13} & f_{23} & f_{33} \end{pmatrix} \\ &=\; \det F \;+\; a_{w_0}\,(f_{22} f_{33} - f_{23}^2) \;+\; a_{w_a}\,(f_{11} f_{33} - f_{13}^2) \;+\; a_{w_0}\, a_{w_a}\, f_{33}. \end{aligned}$$

Each coefficient is a minor of F: $f_{22} f_{33} - f_{23}^2$ is what is left after deleting row and column 1, $f_{11} f_{33} - f_{13}^2$ after deleting row and column 2, and f_{33} after deleting both.

Each term in forecast language. The cofactor formula for an inverse – the adjugate rule behind Cramer's rule, $(F^{-1})_{ii} = (\text{the minor of the } (i,i) \text{ entry}) / \det F$ – names all three coefficients once they are divided by $\det F$:

$$\frac{f_{22} f_{33} - f_{23}^2}{\det F} \;=\; (F^{-1})_{11} \;=\; \sigma_{w_0}^2, \qquad \frac{f_{11} f_{33} - f_{13}^2}{\det F} \;=\; \sigma_{w_a}^2, \qquad \frac{f_{33}}{\det F} \;=\; \frac{1}{\det F_w} \;=\; \det \mathsf{C}_w,$$

the pair's marginalized variances and the determinant of its forecast covariance, for which $\det \mathsf{C}_w = \sigma_{w_0}^2 \sigma_{w_a}^2 - (\mathsf{C}_w)_{12}^2 = \sigma_{w_0}^2 \sigma_{w_a}^2\,(1 - \rho^2)$, with $\rho = (\mathsf{C}_w)_{12} / (\sigma_{w_0} \sigma_{w_a})$ the forecast correlation of w_0 and w_a. In the toy the three numbers are $11/18 = 0.611$, $14/18 = 0.778$ and $3/18 = 1/6$ – the values part two below confirms from the explicit inverse.

The result. Divide the expansion by $\det F$, use the three names, and take the square root:

$$\frac{\mathrm{FoM}'}{\mathrm{FoM}} \;=\; \left[\frac{\det F'_w}{\det F_w}\right]^{1/2} \;=\; \left[\, 1 + a_{w_0} \sigma_{w_0}^2 + a_{w_a} \sigma_{w_a}^2 + (1 - \rho^2)\, a_{w_0} \sigma_{w_0}^2\, a_{w_a} \sigma_{w_a}^2 \,\right]^{1/2}, \qquad (40)$$

the prime marking the noisy forecast, and the relation is exact. Each term $a_j \sigma_j^2 = (\sigma_j / w_j)^2$ is the squared ratio of the parameter's marginalized error to its spurious prior width: the dimensionless measure of how informative the fake prior is. (Careful with the two sigmas: σ_j here is the forecast error of parameter j, not the evaluation noise σ of Table 12; the evaluation

noise lives inside a_j.) In the toy, $\det F' = 18 + 0.5 \times 11 + 0.3 \times 14 + 0.5 \times 0.3 \times 3 = 28.15$, so $\det F'/\det F = 1.564$ and FoM′/FoM $= 1.251$ – the same numbers as the Schur route through F'_w of part one: the two routes agree. And the general case costs nothing new: with m other parameters the linear coefficients become the cofactors of the $(1,1)$ and $(2,2)$ entries, the cross term's f_{33} becomes the complementary minor $\det F_{oo}$, and every step above goes through unchanged.

The first-order rule. When both terms are small, $\sqrt{1+x} \approx 1 + x/2$ and the product term is second order, so

$$\frac{\text{FoM}'}{\text{FoM}} \;\approx\; 1 + \tfrac{1}{2}\, a_{w_0}\sigma^2_{w_0} + \tfrac{1}{2}\, a_{w_a}\sigma^2_{w_a} :$$

one first-order term per noisy column, not one per evaluation – the four evaluations behind a column already sit inside its a_j, through the squared stencil weights $(1+64+64+1)/144 \approx 0.9$ of Eq. (37). With equal terms, $x = a_{w_0}\sigma^2_{w_0} = a_{w_a}\sigma^2_{w_a}$, Eq. (40) is $[1+2x+(1-\rho^2)\,x^2]^{1/2} \approx 1+x$, and exactly $1 + x$ at $\rho = 0$.

First numbers, with the real w_a column. Let only the w_a column be noisy and, for illustration alone, take $\sigma_{w_a} = 1$. At the MCMC settings, $a_{w_a} \approx 0.44$: FoM′/FoM $= (1+0.44)^{1/2} =$ 1.20, a 20% spurious gain. At the final grid, $a_{w_a} \approx 2.8 \times 10^{-4}$: $1 + 1.4 \times 10^{-4}$.

The toy, part two: the forecast errors and the noisy FoM. The forecast errors. The 2×2 inverse of the toy's F_w is

$$\mathsf{C}_w \;=\; F_w^{-1} \;=\; \frac{1}{6}\begin{pmatrix} 11/3 & -10/3 \\ -10/3 & 14/3 \end{pmatrix} \;\approx\; \begin{pmatrix} 0.611 & -0.556 \\ -0.556 & 0.778 \end{pmatrix},$$

so $\sigma_{w_0} = \sqrt{0.611} \approx 0.78$, $\sigma_{w_a} = \sqrt{0.778} \approx 0.88$, and $\rho = -0.556/(0.78 \times 0.88) \approx -0.81$: the strong anticorrelation announced $(1 - \rho^2 \approx 0.35)$. For later use, the full inverse is

$$F^{-1} \;=\; \frac{1}{18}\begin{pmatrix} 11 & -10 & 3 \\ -10 & 14 & -6 \\ 3 & -6 & 9 \end{pmatrix}$$

(its top-left 2×2 block is C_w, as Eq. (39) promises), so $\sigma_o^2 = 9/18 = 0.5$, and o's forecast correlations are 0.30 with w_0 and -0.53 with w_a.

The toy's three cases, collected. Table 14 runs the toy through three noise cases. The first is part one's tens-of-percent additions, $a_{w_0} = 0.5$ and $a_{w_a} = 0.3$: the terms of Eq. (40) are $a_{w_0}\sigma^2_{w_0} = 0.306$, $a_{w_a}\sigma^2_{w_a} = 0.233$ and the product term 0.025, and their sum 1.564 is the same number as $\det F'_w/\det F_w$ from part one's matrices. Once the terms are not small the first-order rule overshoots, and only Eq. (40) itself is exact. The second case shrinks the additions to $a_{w_0} = a_{w_a} = 0.05$: at the few percent the published forecasts care about, first order is already the whole story.

Noise on the other column, the third case. Add $a_o = 0.6$ to the o column alone. This addition sits in f_{33}, inside every subtracted piece of the entry-by-entry formula, so it does not pass through whole; it acts through the marginalization: redoing the Schur complement with $f_{33} = 3.6$ gives $\det F'_w = 6.50$ and the table's ratio $(6.50/6)^{1/2} = 1.041$ – a 4.1% spurious gain from a column that is neither w_0 nor w_a. The mechanism: the fake prior makes o look better measured, a better-measured o absorbs less information, and the pair keeps more. The next paragraph derives the first-order rule that predicts this number from o's correlation with the pair.

noise added	exact FoM′/FoM	first order
$a_{w_0} = 0.5$, $a_{w_a} = 0.3$	1.251	1.269
$a_{w_0} = a_{w_a} = 0.05$	1.034	1.035
$a_o = 0.6$	1.041	1.050

Table 14: The toy's three noise cases: the exact FoM ratio against the first-order rule – Eq. (40) expanded to first order for the pair's columns, Eq. (41) for the o column.

Every noisy column, through its correlation with the pair. This paragraph is the estimate in full, Eq. (41) below: one weight $R_j^2\sigma_j^2$ per column, computed once, times each setting's a_j.

The general case, to first order. In the real system every column is noisy at once: the 2×2 matrix A of the construction grows to the full diagonal, $\mathsf{A} = \mathrm{diag}(a_1, a_2, \dots)$ with one a_j per sampled parameter, and the noisy Fisher matrix is $F + \mathsf{A}$. Near the final settings every $a_j\sigma_j^2$ is small, so first order in A is enough. Three steps: how the inverse moves, which block to keep, and how a determinant responds.

How the inverse moves. For a small A, factor and expand:

$$(F+\mathsf{A})^{-1} \;=\; \big[F\,(I+F^{-1}\mathsf{A})\big]^{-1} \;=\; (I+F^{-1}\mathsf{A})^{-1}\,F^{-1} \;\approx\; (I-F^{-1}\mathsf{A})\,F^{-1} \;=\; F^{-1}-F^{-1}\mathsf{A}\,F^{-1},$$

using $(I+X)^{-1} \approx I - X$ for small X, the matrix version of $1/(1+x) \approx 1-x$. So the change of the inverse is minus the change, sandwiched between two inverses, $\delta(F^{-1}) = -F^{-1}\,\mathsf{A}\,F^{-1}$ – the matrix version of $\delta(1/x) = -\delta x/x^2$.

Which block to keep. The FoM reads the pair's block of F^{-1}. Two pieces of notation first, used to the end of the derivation. Greek indices α, β run over the pair's two parameters (w_0, w_a) only; the Latin index j runs over all sampled parameters. And $(F^{-1})_{w,j}$ is parameter j's column of F^{-1} restricted to the pair's rows – a column of two numbers, entries $(F^{-1})_{\alpha j}$ – while $(F^{-1})_{j,w}$ is j's row of F^{-1} restricted to the pair's columns, the same two numbers laid out as a row (F^{-1} is symmetric); in the toy, for $j = o$, the two numbers are $3/18 = 1/6$ and $-6/18 = -1/3$. Keeping the pair's block of $\delta(F^{-1})$,

$$\delta\mathsf{C}_w \;=\; -\big(F^{-1}\mathsf{A}\,F^{-1}\big)_{ww} \;=\; -\sum_j a_j\,(F^{-1})_{w,j}\,(F^{-1})_{j,w},$$

$$(\delta\mathsf{C}_w)_{\alpha\beta} \;=\; -\sum_j a_j\,(F^{-1})_{\alpha j}\,(F^{-1})_{j\beta},$$

where the sum writes the diagonal A out, column by column: each noisy column lowers C_w by a_j times the outer product of its two-vector with itself (a column times a row, a 2×2 matrix).

How the determinant responds. $\ln \mathrm{FoM} = -\frac{1}{2}\ln\det\mathsf{C}_w$, and for a small change of a matrix S, $\delta\ln\det S = \mathrm{tr}\big(S^{-1}\,\delta S\big)$: the matrix version of $\delta\ln x = \delta x/x$, read off from $\det(I+X) \approx 1 + \mathrm{tr}\,X$ for small X. So

$$\delta\ln\mathrm{FoM} \;=\; -\tfrac{1}{2}\,\mathrm{tr}\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big),$$

with $\delta\mathsf{C}_w$ the block just computed.

From the trace to one number per column. Write the trace out in indices, in two moves. First the matrix product: its (γ,β) entry sums over the shared middle index α (γ is one more Greek index over the pair, since j is taken),

$$\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big)_{\gamma\beta} \;=\; \sum_\alpha (\mathsf{C}_w^{-1})_{\gamma\alpha}\,(\delta\mathsf{C}_w)_{\alpha\beta}.$$

Then the trace: set the two outer indices equal, $\gamma = \beta$, and sum over β – the diagonal entries of the product, added up:

$$\operatorname{tr}\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big) \;=\; \sum_{\beta}\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big)_{\beta\beta} \;=\; \sum_{\alpha\beta}(\mathsf{C}_w^{-1})_{\beta\alpha}\,(\delta\mathsf{C}_w)_{\alpha\beta}.$$

Finally insert the block above with its labels unchanged, $(\delta\mathsf{C}_w)_{\alpha\beta} = -\sum_j a_j\,(F^{-1})_{\alpha j}\,(F^{-1})_{j\beta}$:

$$\operatorname{tr}\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big) \;=\; -\sum_{j} a_j \sum_{\alpha\beta}(\mathsf{C}_w^{-1})_{\beta\alpha}\,(F^{-1})_{\alpha j}\,(F^{-1})_{j\beta}.$$

Each summand is a product of three plain numbers, so it can be reordered at will, $(F^{-1})_{j\beta}\,(\mathsf{C}_w^{-1})_{\beta\alpha}\,(F^{-1})_{\alpha j}$ – a chain $j \to \beta \to \alpha \to j$ in which each index meets its partner in the next factor – and the double sum over β and α is then a row times a matrix times a column, $\sum_{\beta\alpha} x_\beta M_{\beta\alpha}\, y_\alpha = x^{\mathsf{T}} M y$, with x^{T} the row $(F^{-1})_{j,w}$ and y the column $(F^{-1})_{w,j}$: exactly a quadratic form,

$$\operatorname{tr}\big(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w\big) \;=\; -\sum_{j} a_j\,(F^{-1})_{j,w}\,\mathsf{C}_w^{-1}\,(F^{-1})_{w,j},$$

$$\text{and with the } -\tfrac{1}{2} \text{ in front,} \quad \delta\ln\mathrm{FoM} \;=\; +\tfrac{1}{2}\sum_{j} a_j\,(F^{-1})_{j,w}\,\mathsf{C}_w^{-1}\,(F^{-1})_{w,j}.$$

In one sentence: $\delta\mathsf{C}_w$ is a sum of outer products, and the trace of a matrix times an outer product is the matching quadratic form, $\operatorname{tr}(M\,y\,x^{\mathsf{T}}) = x^{\mathsf{T}} M y$. The minus sign of $\delta\mathsf{C}_w$ cancels against the $-\frac{1}{2}$, and the statement is the familiar one: the noise shrinks the forecast ellipse and raises the FoM.

The trace step, checked in the toy. Let only the o column be noisy, with the $a_o = 0.6$ of "Noise on the other column" above. The two-vector is $(F^{-1})_{o,w} = (1/6, -1/3)$, so the outer product gives

$$\delta\mathsf{C}_w \;=\; -\,0.6\begin{pmatrix}1/6\\-1/3\end{pmatrix}\begin{pmatrix}1/6 & -1/3\end{pmatrix} \;=\; \begin{pmatrix}-0.0167 & 0.0333\\ 0.0333 & -0.0667\end{pmatrix}.$$

The trace side, with $\mathsf{C}_w^{-1} = F_w$ from part one: $\operatorname{tr}(F_w\,\delta\mathsf{C}_w) = -0.100$, so $-\frac{1}{2}\operatorname{tr}(\mathsf{C}_w^{-1}\,\delta\mathsf{C}_w) = 0.050$. The quadratic-form side: $(F^{-1})_{o,w}\,\mathsf{C}_w^{-1}\,(F^{-1})_{w,o} = 1/6$ (computed term by term in "R_o^2 in the toy" below, where Eq. (41) splits it as $R_o^2\,\sigma_o^2 = (1/3)(0.5)$), so $\frac{1}{2}\times 0.6\times 1/6 = 0.050$: the two sides agree, as the reordering says they must.

Collected, with each column's number split into named factors,

$$\delta\ln\mathrm{FoM} \;\approx\; \frac{1}{2}\sum_{j} R_j^2\,a_j\sigma_j^2, \qquad R_j^2 \;=\; \frac{(F^{-1})_{j,w}\,\mathsf{C}_w^{-1}\,(F^{-1})_{w,j}}{\sigma_j^2}, \tag{41}$$

one term per column, each the two-column term $\frac{1}{2}\,a_j\sigma_j^2$ weighted by a number R_j^2 between 0 and 1.

What R_j^2 means. It is the squared multiple correlation of p_j with the pair: the share of p_j's forecast variance that a linear fit to (w_0, w_a) explains. It is 1 for w_0 and w_a themselves, recovering the first-order rule of Eq. (40), and 0 for a parameter uncorrelated with the pair: a nuisance column's noise reaches the FoM only through its correlation with the pair. For the

	FoM(w_0, w_a)		FoM$(\Omega_m, 10^9 A_s)$	
column	$k_{\max} = 0.075$	0.2	0.075	0.2
$10^9 A_s$	0.90 (0.87)	0.74 (0.73)	1 (1)	1 (1)
n_s	0.79 (0.74)	0.56 (0.56)	0.97 (0.97)	0.94 (0.94)
H_0	0.76 (0.76)	0.65 (0.65)	0.93 (0.93)	0.89 (0.89)
Ω_b	0.85 (0.77)	0.64 (0.62)	0.98 (0.97)	0.92 (0.92)
Ω_m	0.93 (0.89)	0.84 (0.83)	1 (1)	1 (1)
w_0	1 (1)	1 (1)	0.94 (0.90)	0.88 (0.87)
w_a	1 (1)	1 (1)	0.92 (0.87)	0.82 (0.81)
largest σ_v	0.05 (0.05)	0.12 (0.11)	0.04 (0.04)	0.19 (0.18)

Table 15: The squared multiple correlation R_j^2 of Eq. (41) for each stencil column with the two FoM pairs, on the converged $\Delta z = 0.05$ system, at $k_{\max} = 0.075$ and $0.2\,h$/Mpc ($k_{\max}$ quoted at $z = 0$). A pair's own members have $R_j^2 = 1$ by construction. The last row is the largest of the seven σ_v columns; the other six sit below it. In parentheses: the same R_j^2 computed on the cheap $g = 64$ system; the largest change is 0.08. Computed on the MCMC-settings system instead, the largest change grows to 0.15 (Ω_b at $0.075\,h$/Mpc).

tool, the R_j^2 are the source of the low cost: ranked by their weights $R_j^2\sigma_j^2$, only a few heavy columns contribute, and only their noise needs measuring at each candidate setting.

R_o^2 in the toy. The two-vector is $(F^{-1})_{o,w} = (1/6, -1/3)$ and $\mathsf{C}_w^{-1} = F_w$, so the quadratic form is

$$\begin{pmatrix}1/6 & -1/3\end{pmatrix}\begin{pmatrix}14/3 & 10/3\\ 10/3 & 11/3\end{pmatrix}\begin{pmatrix}1/6\\ -1/3\end{pmatrix} \;=\; \begin{pmatrix}1/6 & -1/3\end{pmatrix}\begin{pmatrix}-1/3\\ -2/3\end{pmatrix} \;=\; -\frac{1}{18}+\frac{2}{9} \;=\; \frac{1}{6},$$

and dividing by $\sigma_o^2 = 1/2$ gives $R_o^2 = 1/3$. No pairwise correlation gives this number – squared, they are $0.30^2 = 0.09$ with w_0 and $0.53^2 = 0.29$ with w_a, and their sum 0.38 overshoots because w_0 and w_a are themselves correlated – which is why the formula needs the full quadratic form. (A check without the formula: with the pair held fixed, o's variance is $1/f_{33} = 1/3$, two thirds of its free variance $\sigma_o^2 = 1/2$; the pair explains the remaining third, $R_o^2 = 1/3$.) The first-order prediction for the toy's last case is then $\delta \ln \mathrm{FoM} = \frac{1}{2} \times \frac{1}{3} \times \frac{1}{2} \times 0.6 = 0.05$, a ratio of about 1.05 against the exact 1.041: the right size, a second-order distance away.

On the real system. Table 15 lists R_j^2 for every stencil column of the converged $\Delta z = 0.05$ system, for both published pairs at both cuts – and, in parentheses, the same numbers computed on a system built from the cheap $g = 64$ columns, for the recipe at the end of this subsection. Every cosmological column is strongly correlated with both pairs ($R_j^2 \approx 0.6$ to 1), so their noise reaches the FoMs almost undiminished, while the seven σ_v columns stay low. The same formula also rates the per-bin columns: the $b_{1,i}$, degenerate with the amplitude A_s, reach $R_j^2 \approx 0.7$ with $(\Omega_m, 10^9 A_s)$ at $k_{\max} = 0.075\,h$/Mpc.

The part that averages to zero. What the average misses. Eq. (41) keeps only the mean of the noise's effect. The middle term of Eq. (36), $2\,d_j^{\mathsf{T}}\mathsf{C}^{-1}e_j$, and the off-diagonal gains $e_j^{\mathsf{T}}\mathsf{C}^{-1}e_k$ have zero mean, but a forecast is run once, on one realization of the computed vectors, and in that one realization they are not zero. So the realized $\delta \ln \mathrm{FoM}$ is the average shift plus a zero-mean remainder, one term per column, linear in that column's noise.

How large the scatter is. Write $X = \sum_j R_j^2\, a_j \sigma_j^2$, twice the average shift of Eq. (41). Two limits bracket the scatter. A column's whitened noise spread evenly over the N_d kept band powers overlaps the few directions the derivatives span only by chance, and the linear terms end with a standard deviation of about $(X/N_d)^{1/2}$ (without priors; priors only lower it). Noise concentrated along the derivative directions is the worst case: one column's linear term can reach the square root of that column's whole contribution to X.

Why the scatter wins near convergence. The average shift is built from the a_j, each proportional to the noise squared; the linear terms grow as the noise itself. Halve the noise and the shift falls fourfold, the scatter only twofold. So close to convergence the zero-mean scatter, not the average push, dominates the realized error.

The practical consequence. A quantity that scatters with the noise realization is not shown converged by shrinking a predicted mean; it is shown converged by checking that the realization does not matter. That is why the practical criterion below is independence of the step – two steps, two different noise realizations, the same FoM – and not the average formula alone: the cheap estimate picks the setting, the step test confirms it.

The prediction against the measured offsets. What the formula is for. Eq. (41) is the convergence diagnostic promised at the top of this subsection: it estimates, cheaply, at which setting the FoM stops changing appreciably – stabilizes. It does not have to predict the FoM's value; the FoMs of the science figures come from the converged derivatives, never from the formula, so order-unit accuracy on the shift is all the diagnostic needs.

What the comparison is. Table 16 compares, for every row of Table 12, the measured FoM ratio of Figure 5 with the ratio predicted from the cheap ingredients alone. Measured: FoM(configuration)/FoM(reference), the grid rows computed with the five-point rule at step $h = 0.02$, the reference the same rule on the finest grid of the starting configuration, $g = 512$. Predicted: F, σ_j and R_j come from the cheap $g = 64$ system (its stencil columns plus the starting configuration's per-bin columns), computed once; each row's noise gives its a_j through Eq. (38), and the prediction is the exact FoM ratio of $F + \mathrm{diag}(a_j)$. The reference is not noise free – its columns carry the same noise floor as the $g = 256$ row, and on the cheap system the formula inflates it by 5.0% at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ and 0.4% at $0.2\,h/\mathrm{Mpc}$ – so the quoted prediction divides each row's inflation by the reference's own. All fourteen stencil columns enter the prediction; the per-bin columns do not – their noise is not measured by Table 12, and swapping them moves the FoM by less than 0.1% ("The distance grid dominates" below), below the comparison's resolution.

How to read the table. Follow $\mathrm{FoM}(w_0, w_a)$ at $0.075\,h/\mathrm{Mpc}$ down the grid rows: the predicted ratio falls $5.98 \to 2.13 \to 1.20 \to 1.00$ while the measured one falls $1.56 \to 1.29 \to 1.006 \to 1.000$. Both settle at 1 – the measurement by $g = 128$, the prediction one doubling later, at $g = 256$. That is the diagnostic doing its job, with a conservative margin: away from convergence the predicted shift is larger than the measured one, so it never declares a setting stable too early. The same reading holds at $0.2\,h/\mathrm{Mpc}$ and for $\mathrm{FoM}(\Omega_m, 10^9 A_s)$. On the final-settings row both ratios lie within a few percent of 1, the stabilized level; the percent-level differences there come from the other controls that change with the final settings, below what the diagnostic is meant to resolve.

Why the margin is conservative. The additions pass whole to F_w only for noise with no preferred direction in data space, while the realized error of the computed vectors is mostly a smooth, broadband shape within each spectrum – the kind of change the nuisance columns

	$k_{\max} = 0.075\,h$/Mpc		$k_{\max} = 0.2\,h$/Mpc		cost at this setting	
configuration	measured	predicted	measured	predicted	full base	cheap (n)
FoM(w_0, w_a)						
MCMC settings	1.556	5.98	1.130	1.935	10 min	35 s (2)
$g = 64$	1.292	2.13	1.059	1.170	29 min	1.7 min (2)
$g = 128$ (starting)	1.006	1.202	1.004	1.024	44 min	3.7 min (3)
$g = 256$	1.000	1.000	1.000	1.000	1.2 h	6.1 min (3)
final settings	1.021	0.969	0.995	0.997	50 min	6.8 min (5)
FoM($\Omega_m, 10^9 A_s$)						
MCMC settings	1.513	2.98	1.115	1.593		
$g = 64$	1.265	1.787	1.049	1.135		
$g = 128$ (starting)	1.007	1.169	1.003	1.020		
$g = 256$	1.000	1.000	1.000	1.000		
final settings	1.018	0.969	0.996	0.997		

Table 16: Measured against predicted FoM ratios on $\Delta z = 0.05$. Measured: FoM(configuration)/FoM(reference) from Figure 5, the reference the $g = 512$ run of the starting configuration (five-point rule at step $h = 0.02$). Predicted, from cheap ingredients only – no converged run enters: the exact diagonal addition $F + \mathrm{diag}(a_j)$, with F, σ_j and R_j from the $g = 64$ system (its stencil columns plus the starting configuration's per-bin columns) and the a_j from the matching row of Table 12, divided by the same prediction for the reference's own noise; the per-bin columns are omitted. The last two columns price the row's numbers with the per-evaluation times of Table 7: the measured side needs a full base at that setting, 154 evaluations, while the cheap prediction needs the fiducial plus four evaluations per heavy column, $1 + 4n$ in all, with n the columns carrying 90% of the prediction, united over both pairs and both cuts. The costs depend only on the setting, so they are given once, in the upper block. One-time costs: the cheap method's $g = 64$ base, $154 \times 11.4\,\mathrm{s} \approx 29$ min, and the measured column's $g = 512$ reference, $154 \times 48.4\,\mathrm{s} \approx 2.1$ h. $k_{\max}$ in h/Mpc, quoted at $z = 0$.

can fit. On the $0.075\,h$/Mpc cut at the MCMC settings, about 95% of the realized w_a-column error lies inside the span of the other columns, and the subtraction of Eq. (39) removes it; the formula counts all of the measured scatter, so its shift is the larger one – the safe direction for a convergence check. (At the MCMC settings the additions are also far beyond first order, $a_{w_a}\sigma^2_{w_a} \approx 30$, Table 13, which is why the predictions use the exact diagonal addition throughout.)

A by-product: an upper bound. Read absolutely, the final-settings prediction bounds the residual noise inflation of the paper's FoMs from above: about 2.6% at $k_{\max} = 0.075\,h$/Mpc and 0.2% at $0.2\,h$/Mpc, for FoM(w_0, w_a) and FoM($\Omega_m, 10^9 A_s$) alike. The offsets themselves are measured, not predicted: the grid scan and the step test below do that.

A cheap check of a setting. The recipe behind the table's predicted column, usable long before any converged run exists. Three steps. First, build one cheap system – here the $g = 64$ columns – and compute F, R_j and σ_j from it, once. Second, for each candidate setting measure the noise of the few heavy columns only: ranked by their contribution $R_j^2\, a_j \sigma_j^2$, two or three columns (w_a, w_0, then Ω_m) carry 90% of the prediction on the grid rows, and five (w_a, Ω_m, H_0, w_0 and one σ_v) at the final settings – the n of Table 16 – four stencil evaluations per column, about 9 to 21 evaluations per candidate setting against 154 for a full derivative base.

In Cocoa time (Table 7), scanning the table's five settings the full way costs about 3.4 h of bases plus the 2.1 h reference; the cheap way, about 19 min of noise measurements plus the 29 min $g = 64$ base. Third, evaluate the prediction and ask where it stops changing by order unity: there the FoM has stabilized – in Table 16, at $g = 256$, one conservative doubling after the measured $g = 128$. The cheap inputs are good enough for that question: recomputed with converged inputs, the predicted ratios move by at most 2% on the near-converged rows and 20% on the coarsest ones, and R_j^2 by at most 0.08 (the parentheses of Table 15) – immaterial for an order-unit diagnostic. And because the predicted shift is the conservative one, a setting the recipe accepts is at least as stable as predicted.

What the estimator does and does not see. The estimate treats the five errors as independent. An error that varies smoothly across the five points, a smooth offset of the whole stencil, cancels in the fourth difference the same way the smooth model does – and it cancels in the finite-difference derivative too, so it does no harm there either. The estimator thus measures exactly the part of the numerical error that hurts a derivative: the part that varies on the scale of the step. Its limit is the in-between case: if the errors at neighbouring points were partly correlated, the differences would partly cancel them, and the estimate would understate the jitter.

At the MCMC settings the Ω_m, w_0 and w_a columns are six to twelve times noisier than the H_0 column, which does not change with the grid. The finer grid quiets w_0 and w_a most: the w_a noise falls fortyfold from the MCMC settings to $g = 256$, the final grid.

How the non-Limber spectra are computed: FFTLog on a distance grid. The exact projection integrates each bin's radial kernel against spherical Bessel functions (Section 6.3; the formulas are in the summary note, `notes/bias_model_summary`, on the non-Limber spectra). The code (`cosmo2D.c`, `C_cl_tomo` and the low-rank core) computes it in three steps:

1. it samples each bin's radial kernels – the density, RSD and magnification rows, and in the low-rank path also the rank kernels – on a log-spaced grid in comoving distance χ covering $z = 0.002$ to 4;

2. FFTLog computes the Bessel transforms $F_\ell(k) = \int d\chi\, W(\chi)\, j_\ell(k\chi)$, and their j_ℓ'' counterparts for the RSD rows: one forward FFT per row, done once, then the Hankel transforms in blocks of 16 multipoles;

3. $C_\ell = (2/\pi) \int dk\, k^2 P(k)\, F_\ell^i F_\ell^j$ is summed on the k grid that FFTLog ties to the χ grid.

Everything therefore rests on how well the χ grid resolves the kernels.

Where the grid factor enters. The grid is endpoint-inclusive in $\log\chi$, with spacing $d\ln\chi = \ln(\chi_{\max}/\chi_{\min})/(N_\chi - 1)$ and $N_\chi - 1 = 511\, g_2$: the combined factor g, the product `accuracyboost` × `nonlimber_accuracyboost`, is rounded up to the power of two g_2 (at most 1024; `init_nonlimber_accuracy_boost` in `generic_interface.cpp`). Doubling g halves every interval and keeps the old nodes, so a higher g strictly refines the same grid, and "about $511\, g$ comoving distances" is exact whenever g is a power of two.

Why thin bins need a fine grid: a worked estimate. The spacing is constant in $\ln\chi$, so a bin of fixed comoving width is narrowest in $\ln\chi$ at high redshift. Take the $\Delta d_{\rm com} = 62$ Mpc layout's ELG bin at $\bar{z} = 1.32$, with a Planck-like cosmology for the estimate: $\chi \approx 4{,}100$ Mpc, so the bin spans $\Delta\ln\chi \approx 62.3/4100 \approx 1.5\times10^{-2}$. Its Fingers-of-God-smoothed edges are far narrower: with $\sigma_v = 289$ km/s (the ELG2 fiducial), the edge width is $\sigma_\chi = (1+z)\,\sigma_v/H(z) \approx 4.6$ Mpc, or $\Delta\ln\chi \approx 1.1\times10^{-3}$. The grid spans $\ln[\chi(4)/\chi(0.002)] \approx 6.7$, so $d\ln\chi = 6.7/(511\,g)$:

- $g = 32$ (the MCMC value): $d\ln\chi \approx 4.1\times10^{-4}$, about 37 nodes across the bin and fewer than 3 across an edge;
- $g = 128$: about 150 across the bin, 11 across an edge;
- $g = 256$ (final): about 290 and 22.

At the MCMC grid an edge is sampled by two or three points. The likelihood records the consequence for the χ^2 itself (`likelihood/desi_cl.yaml`): the smoothed edges of the highest-redshift thin bins resolve between combined factors 32 and 64, and below that the $z = 1.32$ ELG bin carries a coherent 0.18σ broadband amplitude bias.

Why a coarse grid shows up as noise. The grid's endpoints sit at fixed redshifts, $z = 0.002$ and 4, at the sampled cosmology. When a parameter changes the shape of $\chi(z)$, every kernel feature – a bin's edges above all – lands at a different position between the nodes, and with only a few nodes per edge the sampled kernel changes irregularly from one parameter value to the next. That irregular part is the $n(\theta)$ of "The noise of one evaluation". H_0 is the exception that proves the rule: it rescales every distance by one common factor, and on a grid log-spaced between fixed redshifts a common rescaling moves the nodes and the kernels together, so nothing lands differently. That is why the H_0 noise in Table 12 does not change with g, while Ω_m, w_0 and w_a, which change the shape of $\chi(z)$, quiet down as the grid refines.

The distance grid dominates. Table 17 and Figure 5 give the offset of the FoMs from the converged value ($g = 512$), every control but the grid at the starting configuration's value. In the figure, follow the diamonds, the grid scan at step h, from left to right: at $0.075\,h$/Mpc the FoMs fall by 30 to 40% until $g = 128$ and then stay put. The coarse grids add noise that the FoM reads as information. The MCMC settings as a whole (the plus of the figure) raise FoM(w_0, w_a) by 56% at $0.075\,h$/Mpc and by 13% at $0.2\,h$/Mpc, although their fiducial vector differs from the starting configuration's by only $\Delta\chi^2 = 0.015$. The grid alone accounts for most of it.

Which columns carry the offset? Swapping single columns (on the study's system) between configurations shows it: the w_a column, then w_0 and Ω_m. The per-bin and σ_v columns move the FoM by less than 0.1%.

The criterion: independence of the step. The truncation error of a stencil grows with the step, as ϵ^4 for five points, while the noise it divides falls as $1/\epsilon$. A derivative is converged when neither matters: when the FoMs, the errors and the shifts stay put as the step changes, every step scaled together, and as the stencil changes from three points to five.

The step markers of Figure 5, beside each grid, show the test. At $g = 128$ the five-point stencil at $h/2$ raises the FoM by 17%: the noise divided by a smaller step. At $g = 256$, from $h/2$

grid factor g	$k_{\max} = 0.075\,h/\mathrm{Mpc}$	$k_{\max} = 0.2\,h/\mathrm{Mpc}$
32 (MCMC)	+40%	+12%
64	+29%	+5.9%
128 (starting configuration)	+0.6%	+0.4%
256 (final)	0.0%	0.0%

Table 17: The offset of FoM(w_0, w_a) of the paper model from the converged value ($g = 512$), on $\Delta z = 0.05$, every other control at the starting configuration's value. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

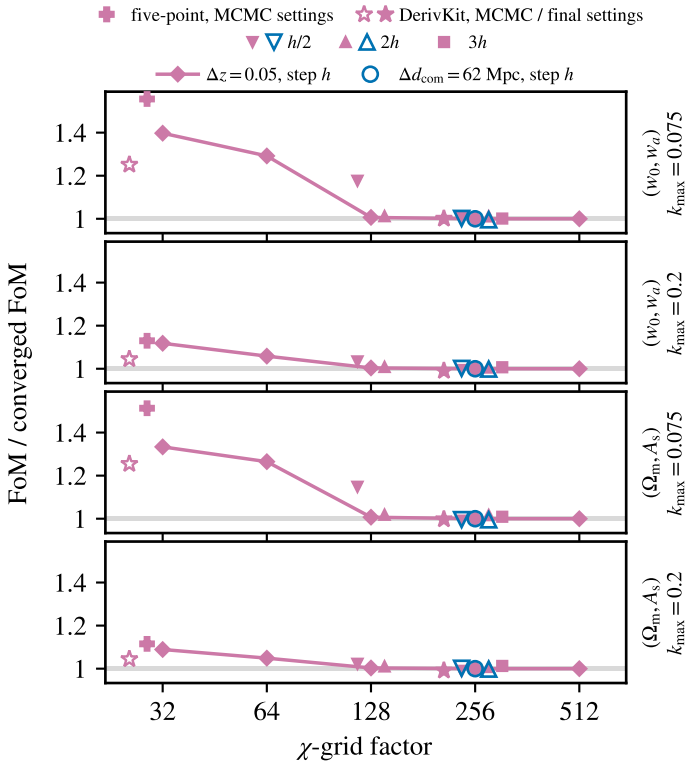


Figure 5: The paper model's FoMs on $\Delta z = 0.05$, as ratios to the converged value (the five-point stencil at step h on the grid $g = 512$). From the top: FoM(w_0, w_a) at $k_{\max} = 0.075$ and $0.2\,h/\mathrm{Mpc}$, then FoM$(\Omega_m, 10^9 A_s)$ at the same cuts. Diamonds: the grid scan at step h, every other control at the starting configuration's value. Down-triangle, up-triangle and square beside each grid: the steps $h/2$, $2h$ and $3h$. Plus: the MCMC settings. Stars: DerivKit, open at the MCMC settings and filled at the final settings. Open blue markers: $\Delta d_{\mathrm{com}} = 62$ Mpc at the final settings, steps $h/2$ and $2h$ against its own step-h value (ring). Gray band: $1 \pm 1\%$. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

to $3h$, both FoMs stay within 1.7%, the shifts move by at most 0.027σ (the study's system), and the three-point stencil agrees with the five-point one. The tolerance on the FoMs is about 1 to 2%; $g = 128$ agrees with $g = 256$ to 0.6% only at $h = 0.02$, so it stays a documented cross-check, and the final tier uses $g = 256$.

The same check on $\Delta d_{\rm com} = 62$ Mpc, at the final settings (the open blue markers of Figure 5, against the ring, that layout's own step-h value): the five-point stencil at $h/2$ and $2h$ stays within 0.6% of step h on both FoMs at both cuts, so $g = 256$ is step independent there too.

CAMB's accuracy. CAMB's `AccuracyBoost` scales its internal precision. The offsets of FoM(w_0, w_a) from the converged value, with FoM$(\Omega_m, 10^9 A_s)$ in parentheses, on the study's system: at 1.05, the MCMC value, the FoM is 14.9% (16.6%) high; at 1.5, 2.0% (2.7%). Raising the boost from the starting configuration's 2 to 3 moves the FoM by 0.2% (0.5%). Every other control moves the FoM by at most 1.9%.

The final settings. The final tier (`DERIVATIVE_OVERRIDES` and `DERIVATIVE_CAMB` of the Fisher module) uses, on every layout:

1. CosmoLike's `accuracyboost` 3, which refines its tables in redshift, wavenumber and multipole, and `integration_accuracy` 1, the number of points of its Limber integrals;
2. the grid factor $g = 256$: `nonlimber_accuracyboost` 64, since $3 \times 64 = 192$ and the code rounds the factor up to the next power of two;
3. the low-rank rank 5 and multipole cap 1000, the values of the yamls;
4. CAMB's `AccuracyBoost` 2.0, `k_per_logint` 30 (wavenumbers per logarithmic interval of its output) and maximum wavenumber $60\,{\rm Mpc}^{-1}$, the likelihood's `kmax_boltzmann` 20 times `accuracyboost` 3;
5. the halofit patch below;
6. five-point stencils for every stencil and per-bin column.

Against the converged value ($g = 512$) on $\Delta z = 0.05$, these settings give FoM(w_0, w_a) +2.1% at $0.075\,h$/Mpc and −0.5% at $0.2\,h$/Mpc, and FoM$(\Omega_m, 10^9 A_s)$ +1.8% and −0.4%. The errors σ move by at most 1.6%, and the largest shift by at most 0.007σ (the study's system).

The remaining 2% sits in the w_0, w_a and Ω_m columns together. At $0.075\,h$/Mpc these three parameters are nearly degenerate, and swapping any one of their columns alone moves the FoM by up to 3.5%, in opposite directions that largely cancel.

DerivKit, the cross-check. DerivKit is an external package, installed by Cocoa under `external_modules/code/derivkit`, that differentiates through a fit rather than a fixed difference. Along one parameter it

1. evaluates the vector at eleven points: the center θ_0 and $\theta_0 \pm 1$, 1.5, 2.25, 3.375 and 5% of θ_0. Below $|\theta_0| = 10^{-3}$ the steps become absolute, 10^{-6} to 4×10^{-5}, far below the noise; that is why the study samples a parameter whose fiducial value is zero (w_a, the b_2

offsets) through the shifted variable $u = 1 + \theta - \theta_0$, whose steps are 0.01 to 0.05 in θ, the range of the five-point steps;

2. fits a straight line through them, entry by entry (`np.polyfit` of degree one for a first derivative), each point weighted by the inverse of its distance to the center. The fit squares the weights, so the center sample carries about a million times the weight of an edge point: the line is pinned to the center;

3. only when some relative misfit exceeds 5%, removes the worst-fitting point and its mirror – never the center – and refits, down to a floor of seven points. The data vector's noise is far below 5% of its values, so the first eleven-point fit is accepted, the pruning never runs, and the fallback `poly_at_floor` is never reached;

4. returns the slope of the line (the calls and defaults of `get_ddv_dkit` in `cosmolike_notebook_utils`, `min_samples` 7).

What the fit really computes. A line pinned to the center sample has a least-squares slope that reduces exactly to the plain average of the ten difference quotients $[m(\theta_i)-m(\theta_0)]/(\theta_i-\theta_0)$, and pairing each point with its mirror turns that into the mean of five central differences, one per step size. This one identity explains the method (Figure 6):

- **Noise.** The center's noise cancels between mirror points, and the average of five independent central differences beats any single one. In the toy of Figure 6, the spread of DerivKit's slope is 1.9% of the true slope against 4.8% for the five-point stencil, a ratio 0.39; the mean-of-five picture predicts the measured spread (1.88% both ways).

- **Bias.** A central difference at step t errs by $m''' t^2/6$ (the even orders cancel between mirror points), so the mean keeps $(m'''/6)\,\langle t^2\rangle$: a curvature bias the five-point stencil cancels by construction. In the toy it is +0.5%, the offset of the noise-free marker.

- **What no fit removes.** A smooth deterministic error, the signature of a coarse grid, moves all eleven points together: a bias, not a scatter. The toy's smooth ripple deflects all three estimators by the same 3.1 to 3.2%. A fit averages scatter; it cannot remove a bias.

The study applied DerivKit to the cosmological and σ_v columns and to the $\{b_{1,i}\}$ and $\{b_{2,i}\}$ families, batch by batch (the stars of Figure 5).

- At the MCMC settings DerivKit's FoM is 25% high, against 56% for the five-point stencil: it reduces the noise but cannot remove the error of a coarse grid.

- At the final settings it agrees with the five-point stencil within 2.0% (σ within 2.2% and shifts within 0.006σ on the study's system) and sits within 1% of the converged value ($g = 512$).

- It costs 11 evaluations per column against 4, about 210 s against 78 s per column.

The final tier therefore uses five-point stencils everywhere, with DerivKit as the documented cross-check.

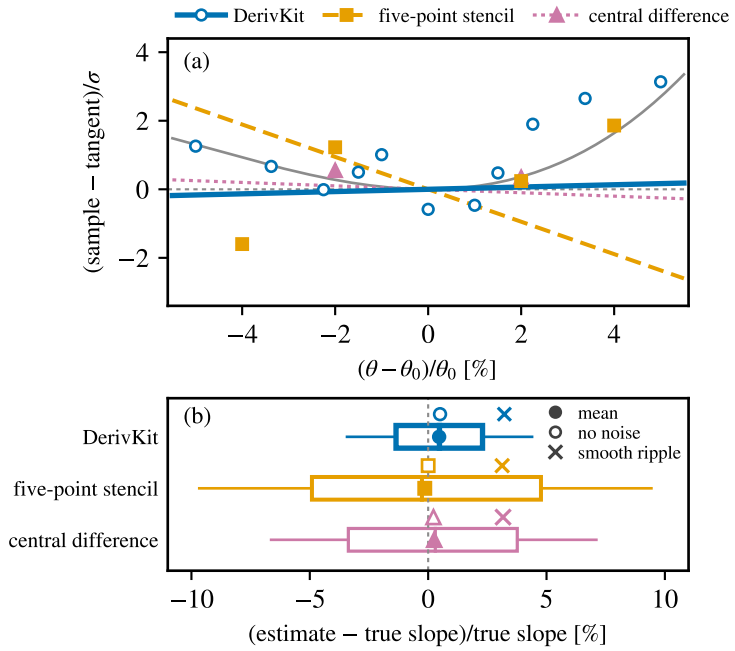


Figure 6: A toy comparison of the three derivative estimators, not a measurement: $m(\theta) = 1 + t + 0.8\,t^2 + 5.6\,t^3$ with $t = \theta - \theta_0$ and $\theta_0 = 1$, so the true slope is 1; independent Gaussian noise of $\sigma = 10^{-3}$ at every evaluation; and the estimators called as the study calls them: a central difference at $\pm 2\%$ of θ_0, the five-point stencil at ± 2 and $\pm 4\%$, and DerivKit at the center and ± 1, 1.5, 2.25, 3.375 and 5%. (a) One typical realization: each estimator's samples minus the true tangent at θ_0, in units of σ, with its fitted slope drawn through the center; the gray curve is the true curve minus the tangent. (b) The error of the estimated slope over 2,000 realizations: the box spans the 16th to 84th percentiles, the whiskers the 2.5th to 97.5th, the filled marker is the mean, the open marker the noise-free estimate (the curvature bias), and the cross the error on a smooth deterministic ripple, $10^{-3}\sin(2\pi t/0.2)$, added without noise, which every estimator follows alike.

The CAMB halofit patch. Halofit is CAMB's fitting formula for the nonlinear matter power spectrum. It finds the nonlinear scale by bisection (repeated halving of an interval) on $\sigma(R)$, the rms linear density contrast smoothed on the scale R; stock CAMB stops at $|\sigma(R) - 1| \leq 10^{-3}$. Where the bisection stops jumps between nearby cosmologies and shifts the nonlinear power spectrum by about 0.3%, a noise that a finite difference divides by its step.

Cocoa's installation key `PATCH_CAMB_HALOFIT_TOLERANCE`, on by default, tightens the test to 10^{-7}. The final tier requires it: the builders stop at start unless `halofit.f90` holds the 10^{-7} test and the compiled CAMB library is newer than that file (`check_halofit_patch`).

Two tiers. The project fixes two sets of accuracy settings, two tiers, which `derivative_settings` returns per layout.

- **Exploration**: fast settings for quick looks; its FoMs are not converged.
- **Final**: the settings above, for every figure and number of a paper or of a note's final

version. The criterion is the stability of the FoMs, the errors and the shifts; runtime is not a criterion.

Both tiers use five-point stencils for every stencil and per-bin column, and exact templates for $\{\mathrm{SN}_{0,i}\}$ and $\{\mathrm{SN}_{2,i}\}$. A cache made with central differences belongs to neither tier, and the builders refuse it.

Each cache records its tier, its settings, and a fingerprint (`settings_fingerprint`): a 64-character code (a sha256 hash) computed from the full model input, so that any change of the input changes the code. The input is the example yaml merged with the likelihood's options and parameter files, the accuracy settings, the priors and the ties between parameters, together with the fiducial point and the tier. Unlike a comparison of file names, the fingerprint changes when a yaml is edited under the same name, and stays the same when a file only moves.

The two routes agree. The study computes its derivatives by a second route: it unties the per-bin parameters and steps them through the full likelihood, with no step class (Appendix A). The agreement study (`tests/studies/results/fisher_derivatives_agreement_dz0p05.json`) compares that route with the production cache of $\Delta z = 0.05$ at the final tier, column by column. The per-bin $b_{2,i}$, $b_{s^2,i}$ and $\alpha_{0,i}$ columns agree exactly (relative difference zero); the $b_{1,i}$ columns, which the two routes connect through the chain rule of Eq. (13), agree to 5.2×10^{-11}; and the cosmological and σ_v stencil columns to 4.6×10^{-14}.

10 Limits and checks

What needs new evaluations of the data vector.

- **A physical term outside the base.** A term that is no combination of the knobs, such as a nonzero third-order bias or redshift-space one-loop terms (which CosmoLike lacks), needs new base columns. A new combination of existing knobs, such as a quadratic tilt or another calibration slope at the same fiducial values, is only a new M.

- **A different fiducial point.** Every derivative is taken at p_0. A new fiducial cosmology, new fiducial biases, or a calibration law that moves the fiducial $b_{2,i}$ needs new base columns, and a new covariance, which sits at the one-loop fiducial. A new covariance alone needs no evaluation: the notebook rebuilds every G_K from the current covariance file at every run.

- **Dependence beyond first order.** The forecast is the curvature at p_0, and Eq. (25) is first order in δm. The band powers are quadratic in the biases, so a large truth, such as an offset of twice a prior width or the 22% BGS flux-limit drift (the change of the BGS linear bias from one end of the group's redshift range to the other that the sample's flux limit produces, about 11% at either edge), is treated to first order, and a shift near the $\sigma/3$ threshold (a third of the parameter's forecast error) needs a refit, with new evaluations, to be confirmed.

The checks that guard the machinery.

- **The chain rule.** The per-bin columns, obtained through the step classes and summed over a group, $\sum_{i \in g} \partial m / \partial \beta_i$ with Eq. (13), must equal the stencil derivative $\partial m / \partial B_g$,

obtained through the full likelihood. The two routes share no evaluation; the notebook's own check finds they agree to at most 1.5×10^{-9} of the stencil column's largest entry on every layout (the truncation error of the two stencils), and the final-tier agreement study of Section 9 tightens that to 5.2×10^{-11} on $\Delta z = 0.05$.

- **Cross-cache checks.** The order-0 templates of the template cache equal the derivative cache's $\mathrm{SN}_{0,i}$ templates, and the two caches' fiducial vectors are equal. B carries the difference of the two vectors as its check column, so their $\Delta\chi^2$ is a diagonal entry of G on the shipped mask; it is zero on every layout. The vector after the last step equals the fiducial one bit for bit (the `reset` entry): no step survives its evaluation.

- **The matched-prior identity.** When the truth's linear-bias scatter equals the fit's prior and every other parameter is drawn from the fit's own priors, the scatter and noise terms of Eq. (26) must add up to exactly n_{dof}: the widening the prior gives the ellipse equals the error the scatter causes (the correlated-prior note, on the test statistic).

- **One answer from independent constructions.** A + and a − truth give shifts of equal size (measured 0). Section 9 of the notebook builds the paper model's M by its own route, and at σ_v width factor 1 it reproduces the α_2-fixed fit of section 11 exactly; the pair-threshold scan with every pair kept reproduces the FoMs of sections 6 and 9 exactly. Each agreement tests G, M, the priors and the pseudo-inverse together.

- **The cache check.** Each cache stores its settings, tier and fingerprint, and the notebook refuses a cache whose entries differ from its own (Section 11).

11 Using the caches

The caches live in two places. The working copies, which the notebooks read and write, are the files of Table 11, one set per layout, in `chains/`, which git ignores. Final-tier copies are archived in `fisher_caches/`, whose `.npz` files git stores through Git LFS, so a fresh clone copies an archived cache instead of recomputing it (hours). A cache's name says what it holds, the layout and the tier, with the layouts `dcom62` ($\Delta d_{\mathrm{com}} = 62$ Mpc), `dz0p02`, `dz0p05` and `dz0p20` ($\Delta z = 0.02$, 0.05, 0.2).

- `model_scale_cuts_final`, written by the notebook's section 3: the bands each ultraviolet-fraction cut keeps, computed on the paper fiducial at the final derivative settings (those of the derivative base), at $k_{\max} = 0.05, 0.075, \ldots, 0.25\,h/\mathrm{Mpc}$, with its settings and fingerprints; the scans use the seven cuts up to 0.2. (The linear notebook, `EXAMPLE_EVALUATE_FISHER.ipynb`, keeps its own cut cache on the linear-bias fiducial, `linear_scale_cuts_final`, for the linear-bias experiments.)

- `model_derivatives_<layout>_final` and `model_templates_<layout>_final`: the base caches of Table 11 – the stencil and per-bin operator columns, the $\mathrm{SN}_{0,i}$ templates, the fiducial vector and biases in the first; the exact α_2 columns and the $\mathrm{SN}_{0,i}$ and $\mathrm{SN}_{2,i}$ templates in the second – written by the notebook's section 4 or by `scripts/compute_fisher_base_mpi.py`, which writes the same files bit for bit.

Neither the notebook nor the script caches a Fisher matrix, G_K or FoM: these depend on the covariance and are rebuilt at every run.

The file format. A cache is one `.npz` file (a zip archive of numpy arrays) of plain arrays plus one JSON header. A pickle can run code when it is loaded; plain arrays and JSON text cannot, so reading a cache runs no code. The header lists the settings, the tier, the fingerprints, the parameter names, the shape of every array and the date, and it holds a provenance record: for every model the fully resolved cobaya input with its fingerprint, the sha256 of every dataset file, the stored settings, and the git state of the repositories, the CAMB version and the halofit patch at write time. The archive keeps the same record next to each file as `fisher_caches/<name>.json`, in plain git, readable on GitHub, and `fisher_caches/checksums.json` holds the sha256 of every archived file.

Four calls, each explicit in its own cell. Nothing loads, computes or writes a cache on its own; no code checks whether a file exists to decide what to do. A notebook cell calls one of four functions of the Fisher module, so every load and every save shows intent, and a changed setting can never reuse or overwrite a cache unnoticed.

- `load_fisher_from_cache` copies an archived cache to its working copy. It refuses a Git LFS pointer (the small text file git leaves for a large file it never downloaded; the message names `git lfs pull`), a checksum that differs from `checksums.json`, and a different existing working copy without `overwrite=True`.
- `read_fisher_cache` reads a working copy. It raises when the stored settings differ from the notebook's, when a tier entry differs from the notebook's tier, or when a stored fingerprint no longer matches the current yaml, parameter and dataset files; it never recomputes.
- `write_fisher_cache` writes a freshly computed result, with its settings and the provenance record, to the working copy. It refuses an existing file without `overwrite=True`, and writes nothing when a stored fingerprint differs from the current files.
- `save_fisher_to_cache` archives a working copy, records its checksum and writes the record `.json`: the only call that changes the archive. Only final-tier results enter the archive; it refuses a cache without a tier or with any tier other than final, a changed fingerprint, and a different archived file without `overwrite=True`.

The compute cells are committed commented out: to recompute, run a compute cell in place of its read cell, then the write cell; the layouts are computed cheapest first. The MPI script follows the same rules with `--overwrite`, writes working copies only, and stops without writing when its ranks see different files or a file changes during the run.

What makes a cache out of date, so it must be recomputed.

- **Detected automatically**: any change of the model input that the fingerprint covers (a likelihood option, a CAMB argument, a prior, a tie between parameters, the fiducial point), of the tier, of the stencil rule or of the steps. The read cell stops and names the mismatch. A file that moves or is renamed with the same content keeps its cache.
- **Not detected, so recompute by hand**: a change of the code itself (the likelihood's Python, CosmoLike, CAMB beyond the halofit check), since the fingerprint covers the

input, not the code. No check compares a cache with the current code unless a cache is recomputed; the fiducial-vector check then catches a change that moves the fiducial vector.

- **Needs no recomputation**: a new covariance, a new prior or prior width, a new model built from the cached knobs, and any cut already in the cut cache. A $k_{\max}$ outside it needs the cut compute cell for that cut (CosmoLike's split of each band into the power below and above the cut), but no new derivative.

A The per-bin step classes

Why a step needs the likelihood's arrays. The sampled parameters offer no per-bin handle: the parameter files tie each bin's linear bias to its group amplitude, and the likelihood computes $b_{2,i}$ and $b_{s^2,i}$ from $b_{1,i}$ through the calibrations, recomputing and resending every per-bin value at every evaluation. The completed per-bin values appear in one place: the argument lists of the compiled setters through which the likelihood hands them to CosmoLike (`set_nuisance_bias_tidal` for $b_{1,i}$, $b_{2,i}$, $b_{s^2,i}$ and the slot that carries $\alpha_{0,i}$; `set_nuisance_bias_alpha2` for $\alpha_{2,i}$). The batch steps of Section 6.2 are applied there. Resending the unchanged values does no harm; a step only has to change the entries of the batch's bins and leave everything else as it is.

A subclass of the likelihood. The notebook and the MPI script evaluate each layout on a subclass of its paper likelihood, defined in `scripts/compute_fisher_base_mpi_classes.py`: the mixin `PerBinStepMixin` listed before the likelihood class in the subclass's bases (for $\Delta d_{\rm com} = 62$ Mpc, `desi_cl_model_b_steps`). A mixin is a class never used alone: its methods run first and reach the likelihood's own through `super()`. It adds no option and no parameter – only three methods a caller uses:

```
like.set_per_bin_step(operator='b2', bins=members, step=0.02)
vector = model.dv()        # the point with the step added
like.clear_per_bin_step()  # the next point is the likelihood's own
```

`operator` is one of `b1`, `b2`, `bs2`, `a0` and `a2`, `bins` the 0-based bins of the batch, and `step` the signed step. The set and the clear are wrapped in `try/finally` (`evaluate_with_step` in the notebook), so no error leaves a step installed. Without a step the subclass evaluates exactly what its parent does.

The stepped evaluation. The likelihood's `set_lens_related` computes a point's per-bin arrays and sends them to CosmoLike. The mixin overrides it: the parent runs in full – every array, the Fingers-of-God $n(z)$, every setter – and then, with a step set, the stepped operator's setter is called once more with the step added (trimmed):

```
def set_lens_related(self, **params):
    super().set_lens_related(**params)   # the point's own arrays
    if self._step is None:
        return
    # fresh copies of the arrays the parent just sent
    B1, B2, BS2, BK = ...
    if operator == 'b1':
```

```
        B1[members] *= 1.0 + step
    elif operator == 'b2':
        B2[members] += step
    elif operator == 'bs2':
        BS2[members] += step
    elif operator == 'a0':
        # alpha_0 k^2 P = 2 b1 bK k^2 P with k in H0/c units
        BK[members] += step/(2.0*B1[members]*C_OVER_H0**2)
    ci.set_nuisance_bias_tidal(B1=B1, B2=B2, BS2=BS2, ..., BK=BK)
```

`B1`, `B2` and `BS2` hold $b_{1,i}$, $b_{2,i}$ and $b_{s^2,i}$, and `BK` the nonlocal bias $b_{K,i}$, one entry per bin. The $b_{1,i}$ step is relative, the $b_{2,i}$ and $b_{s^2,i}$ steps absolute, and an $\alpha_{0,i}$ step enters `BK`, because CosmoLike writes the counterterm of Eq. (3) as $2\,b_{1,i}b_{K,i}\,k^2P_{\rm mm}$ with k in units of H_0/c (`C_OVER_H0` $= c/H_0$ in Mpc/h). An `a2` step instead calls `set_nuisance_bias_alpha2` with the step added to the point's $\alpha_{2,i}$ on the batch's bins. Each compiled setter stores a value only when it differs from the stored one by more than a relative 10^{-13}, then renews the cache key of the bias tables and computes nothing else, so the second call leaves CosmoLike in the state that one call with the shifted arrays leaves it in.

The model stays the paper yaml's. Cobaya finds the subclass through the `class` and `python_path` keys of a likelihood block. `step_yaml` writes them into a temporary copy of the paper yaml; a cobaya class without a yaml file of its own inherits its parent's defaults (options, parameter files, dataset), so every model detail still comes from the parent. `check_step_yaml` stops unless the copy resolves to the same cobaya input as the paper yaml – the likelihood options, theory, parameters, priors and fiducial point, compared blockwise.

The batch loop. The loop of the notebook's `fisher_base` walks the four stenciled families and the four batches (trimmed; `OPERATOR_STEPS` holds the steps ϵ of Section 6.2: 0.01 for $b_{1,i}$, 0.02 for $b_{2,i}$ and $b_{s^2,i}$, $5\,h^{-2}\,\mathrm{Mpc}^2$ for $\alpha_{0,i}$):

```
for op, step in OPERATOR_STEPS.items():
    Dop = np.zeros((len(m0), nbins))
    for c in range(max(color) + 1):
        members = np.array([i for i in range(nbins) if color[i] == c])
        vectors = []
        for offset in fm.FIVE_POINT_OFFSETS:
            vectors.append(
                evaluate_with_step(model, like, op, members, offset*step))
        d = fm.five_point(vectors, step)
        for i in members:
            # the b1 step is relative: (b1 dm/db1)/b1 = dm/db1
            Dop[rows[i], i] = d[rows[i]]/(b1[i] if op == 'b1' else 1.0)
    ops[op] = Dop
```

`color[i]` is bin i's batch number from the greedy rule of `pair_coloring` (the code keeps the mathematicians' word for a batch); `fm.FIVE_POINT_OFFSETS` $= (2, 1, -1, -2)$ are the offsets o of Eq. (14), and `fm.five_point` combines the four vectors with the stencil weights and divides by the step. `rows[i]`, built once by `spectrum_rows`, lists the entries of every spectrum that contains bin i; column i receives the derivative on those rows and stays exactly zero elsewhere, so the zeros of Table 3 are never computed. The $b_{1,i}$ column is divided by $b_{1,i}$,

because its relative step makes the stencil return $b_{1,i}\,\partial m/\partial b_{1,i}$. The α_2 loop is the same with two evaluations per batch (Section 6.3). The MPI script evaluates the same points with the same classes and combines them with the same arithmetic; one rank runs them in this order.

The caches. Every stencil evaluation is at the same sampled point; the step rides inside the likelihood. Cobaya (the framework that runs the likelihood) stores the previous point's result and would return it without calling the likelihood, so `ModelFisher.dv` (`interface/cosmolike_lssty1xdesi_fisher.py`) calls `self.model.loglike(point, ..., cached=False)`, and the likelihood, with its step, runs at every evaluation.

CosmoLike keeps caches of its own, one key per sector of inputs (`cosmolike_core`'s `README.md`, section "The sector-ladder cache check"). A setter compares every incoming value with the stored one (`fdiff` in `generic_interface.cpp`) and, when any differs, draws a fresh random number as the sector's key (`nuisance.random_galaxy_bias` for the biases); every cached projection stores the keys it was built under and recomputes on a mismatch (`cosmo2D.c`). A batch evaluation therefore rebuilds the bias-dependent results and reuses everything keyed on other sectors – CAMB's power spectra, the distances, the power and FAST-PT tables. On $\Delta d_{\rm com} = 62$ Mpc at the final settings, an evaluation whose only change is the biases takes about 36 s, against about 2.5 minutes at a new cosmology (the laptop of Section 6.4). The return to the fiducial values is one more change, caught by the same comparison, and clearing the step needs no cleanup: the likelihood resends every per-bin value at the next point anyway.

What tests this. The sector-key machinery is tested by the sector-ladder cache check, `tests/data_vector/test_cache_consistency.py` (`tests/data_vector/README.md`, section "The sector-ladder cache check"): a deterministic ladder of mixed cosmology, bias and Fingers-of-God updates on all four layouts, a scramble of every sector at once, and bitwise assertions, with a second model walking the mirrored ladder. That test runs the frozen linear-bias configuration, so it exercises the linear setter `set_nuisance_bias`, not the one-loop `set_nuisance_bias_tidal` (Appendix B). The stepped route itself is validated against derivatives that never pass through a step class: the notebook's chain-rule check reproduces the group-amplitude stencils from the per-bin columns through Eq. (13), to at most 1.5×10^{-9} of the group column's largest entry over the four layouts, and the route-agreement study of Section 9 unties every per-bin parameter and steps it through the full likelihood: its $b_{2,i}$, $b_{s^2,i}$ and $\alpha_{0,i}$ columns agree exactly with the stepped ones, the $b_{1,i}$ columns through the chain rule to 5.2×10^{-11}. The reset check of Section 10 confirms, bit for bit, that no step survives its evaluation.

B Open issues

- **The sector-ladder test has no one-loop rung.** The cache-consistency ladder exercises the linear setter `set_nuisance_bias` only; a ladder whose bias updates run through `set_nuisance_bias_tidal` (offset, $\alpha_{0,i}$ and $\mathrm{SN}_{0,i}$ steps) is still to be added.

- **Some accuracy offsets are scored on the study's system.** The error and shift stabilities of Section 9 (the 0.027σ step stability, the 1.6% of the errors and 0.007σ at the final settings, DerivKit's 2.2% and 0.006σ), the CAMB-boost and other-control offsets and the column-swap attribution are scored on the fiducial model with α_2 free and the

exploration σ_v widths, the convergence study's own system; only the FoMs were rescored for the paper model on the paper-fiducial cuts. The derivative columns are shared, so a rescoring would change the quoted stabilities, not the chosen settings.

- **The extra cross pairs, and DSATUR.** The pairs of bins two apart do carry information: `notes/cross_correlations` forecasts the extended lists (new pair lists, masks, covariances, scale cuts and bases, cached next to the current caches; Section 6.5) and attributes the gain to the free bias levels and slopes rather than to line-of-sight information of their own. The analysis keeps the current list; the extended lists are a comparison against it. The code keeps the greedy coloring in bin order because it is simple to explain and to check; on those extended lists it uses six batches where four suffice, 18 extra evaluations per batch, an accepted cost. DSATUR, which reaches the minimum in every case, stays an open item for a future study.

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